

Fear as a Barrier to Coverage: The Chilling Effect of the Public Charge Rule on Children's Medicaid Enrollment

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Abstract

Objective: To examine whether the Trump administration's 2019 public charge rule, which made Medicaid receipt a negative factor in certain immigration determinations, caused coverage losses among populations exempt from the rule due to fear. There was significant concern that the rule would cause immigrants to avoid health coverage - a "chilling effect" - even though the rule did not legally apply to most.

Data Sources: American Community Survey (ACS) microdata, 2014-2024 (N = 2,077,869 Medicaid-eligible U.S. citizen children below 200% of the federal poverty level); restricted-use California Health Interview Survey (CHIS) adult files, 2015-2024 (N = 220,925 adults).

Study Design: Difference-in-differences and event study models comparing citizen children in mixed-status families with children in citizen-only families; four-level decomposition of citizenship status in California (U.S.-born, naturalized, green-card holder, no green card).

Principal Findings: In the national analysis, citizen children in mixed-status families saw a 2.0 percentage point decline in Medicaid enrollment after the September 2018 public charge announcement ($p < 0.001$). The decline has persisted through the policy reversal period and deepened during the 2023-2024 Medicaid redetermination restart (the "unwinding"). Declines were primarily among families with a noncitizen parent: decomposing the citizen-only control group shows that the chilling effect concentrates in these households (-1.8 pp Medicaid relative

to the U.S.-born-parent baseline, $p = 0.003$) rather than across immigrant-connected families broadly (naturalized-parent x post: +1.4 pp, $p = 0.06$). Declines were also sharper outside immigrant-inclusive states: excluding seven jurisdictions that extended Medicaid or near-equivalent state-funded coverage to undocumented adults during the paper window (California, Illinois, Oregon, New York, Colorado, Washington, and the District of Columbia) increases the chilling estimate to -2.6 pp ($p < 0.001$), consistent with inclusive state policy environments partially offsetting the federal rule. In California, naturalized citizens saw a 1.3 percentage point Medi-Cal decline ($p = 0.030$) and effects were larger for immigrants with lesser citizenship status: adults without green cards are 20.5 percentage points more likely than naturalized citizens to report having declined a non-cash benefit because of immigration concerns ($p < 10^{-30}$), a steep gradient for the households that make enrollment decisions that may spill over to children. California's recent full-scope Medi-Cal expansions for undocumented adults (January 2020 ages 19-25, May 2022 ages 50+, January 2024 ages 26-49) make precise estimation of coverage-chilling difficult, but suggest a blunting effect: in cleanest within-state cell (age 26-49 in 2019-2021), Medi-Cal coverage among adults without green cards rose +6.8 pp rather than fell.

Conclusions: The public charge rule resulted in coverage losses among eligible citizen children who were not legally affected by the rule, and concentrated in mixed-status families and in non-expansion states. Supplemental California adult evidence provides support that these spillovers are widespread among the immigrant community and are based in fear-based avoidance and administrative vulnerability.

1. Introduction

1.1. Motivation

The Trump administration's 2019 expansion of the public charge rule redefined what counted as a "public charge" for immigration purposes to include receipt of Medicaid, the Supplemental Nutrition Assistance Program (SNAP), and federal housing assistance. While the rule technically applied only to certain noncitizens applying for admission or adjustment of status, its early-2017 leak and September 2018 announcement triggered fear, confusion, and benefit avoidance that extended well beyond the populations it formally targeted. The overarching goal of this paper is to examine whether the public charge rule reduced Medicaid enrollment among legally exempt populations, and through what channels (fear, administrative friction, or information failure) a policy directed at immigration status translated into reduced health coverage for U.S. citizen children and other formally exempt groups – as well as provide a causal estimate for the "chilling effect", which so far has only been estimated through surveys.

This fear can now be measured directly rather than through hypothetical means. Beginning in 2019, the California Health Interview Survey asked adults directly whether they had ever avoided a non-cash government benefit because they worried it would jeopardize a green card or citizenship application. Avoidance rises steeply with documentation status, and it is steepest among adults raising children (Section 5.8, Table 6).

That premise motivates the central empirical question of this paper: If fear of this size circulates through immigrant communities, does it translate into measurable coverage losses among the populations the rule never touched - particularly U.S. citizen children in mixed-status families, who are eligible for Medicaid on exactly the same terms as any other citizen children? By

analyzing national and state survey data, I am able to provide causal evidence on coverage effects and elaborate on effects of the fear gradient.

This question is a matter of substance for many reasons, first and foremost being children's healthcare coverage. Medicaid is the primary source of health coverage for low-income children in the United States - in 2018, Medicaid and CHIP covered approximately 37 million children, nearly half of all children below 200% of the federal poverty level (FPL), and decades of research establish that childhood Medicaid coverage improves health outcomes, educational attainment, and long-run economic self-sufficiency (Currie and Gruber, 1996; Cohodes et al., 2016; Wherry and Meyer, 2016; Miller and Wherry, 2019). Any policy that reduces enrollment among this population has consequences that compound over a child's lifetime. This analysis also exposes a structural vulnerability in the American safety net - recent estimates place the mixed-status household population near 22 million people, and close to 5.5 million U.S.-citizen children live with at least one undocumented household member (FWD.me, 2024). Their noncitizen parents, however, face real or perceived immigration risk from interacting with government enrollment systems. When immigration policy becomes more punitive, parents in mixed-status families face a decision that parents in citizen-only families do not: whether enrolling a child in an entitled program might jeopardize a parent's ability to remain in the country, or even the child's own future in some way.

In addition, public charge has re-entered the public spotlight. The second Trump administration has signaled renewed interest in expanding public charge determinations – with a final public charge rule being released in July 2026 - and immigrant communities are again reporting elevated fear and benefit avoidance (Bernstein, Gonzalez, and Guelespe, 2025). Understanding the earlier rule's chilling effect and long-term persistence after the rule's reversal is therefore

directly relevant to current policy debates. If effects endure long after a restrictive rule is rescinded, the costs of such policies are substantially larger than what the formal provisions alone would suggest. The setting is also useful for the broader take-up literature, as a large body of work documents that program participation is shaped by transaction costs, stigma, information barriers, and perceived risks of government interaction (Moffitt, 1983; Currie, 2006; Bhargava and Manoli, 2015; Herd and Moynihan, 2018). What makes this analysis particularly unique in relation to the literature is that public charge rule changed none of the formal eligibility rules for the populations this paper studies - no Medicaid income thresholds, categorical eligibility, or benefit packages were altered – the only thing that changed was the perceived cost of participation and the visibility of immigration risk in the enrollment process, which makes the episode closer to a natural experiment in the effects of fear and administrative burden on take-up.

1.2. What this paper does

The paper has two empirical components. The primary analysis uses ACS microdata on Medicaid-eligible U.S. citizen children from 2014 to 2024 to estimate difference-in-differences and event-study models comparing children in mixed-status families with those in citizen-only families. The ACS provides a nationally representative sample large enough to detect econometrically meaningful coverage changes, as well as individual-level data on household immigration composition and coverage spanning the full public charge policy cycle from pre-announcement through the Medicaid unwinding, with four pre-announcement survey years available for pre-trend testing.

The supporting secondary California analysis uses restricted-use CHIS microdata from 2015 to 2024 to compare adults across four citizenship groups: U.S.-born citizens, naturalized citizens, noncitizens with green cards, and noncitizens without green cards. This data provides an

opportunity for a more granular, nuanced estimation of chilling that considers documentation status within the noncitizen population, as restricted-use CHIS distinguishes green card holders from other noncitizens; because green-card holders and undocumented adults face widely different public charge exposure, this segmentation provides more nuance than public-use CHIS allows. While CHIS cannot exactly reproduce the ACS child estimate inside one state, it adds several outcomes the ACS cannot observe: Medi-Cal enrollment for naturalized citizens, emergency department use, usual source of care, forgone care, and whether respondents report being asked for a Social Security number or citizenship documentation when seeking coverage. The restricted-use file also includes two fear-avoidance items collected from 2019 onward (lifetime and 12-month recall of deciding not to apply for a non-cash benefit over green-card or citizenship concerns).

The two analyses play different and complementary roles. The ACS estimates identify the main child-coverage effect in the population for whom Medicaid is the relevant margin; as the California data is from a singular state with outlier immigration policies, I frame its results as secondary and supporting rather than central. Regardless, the CHIS analysis adds three complementary components: (1) It tests for chilling spillovers to fully exempt populations - naturalized citizens were unambiguously not subject to the rule, yet remained embedded in immigrant community fear networks, (2) it identifies a direct fear-based benefit-avoidance gradient using post-period data, which is a precisely measured version of the channel through which public charge fears translate into foregone enrollment, and (3) it provides an example of how individual state responses can significantly affect chilling and accuracy of measurement: the contemporaneous full-scope Medi-Cal expansions for undocumented adults preclude a clean within-state coverage-chilling estimate, but CHIS decomposition allows some separation of

federal chilling from state-expansion effects and the cleanly identified spillover, fear-avoidance, and administrative-vulnerability gradients.

1.3. Contribution

I add to the literature in several ways: First, the paper provides individual-level quasi-experimental evidence consistent with a chilling effect of the public charge rule on Medicaid enrollment among low-income citizen children in mixed-status families. The preferred estimate (a 2.0 percentage point decline, with event-study leads that are jointly indistinguishable from zero across four pre-announcement years) is consistent with prior county-level estimates (Barofsky et al., 2020) but extends the evidence to individual-level microdata, household-level treatment definition, and the full 2014-2024 policy cycle (Section 2.5).

Second, the paper shows that the child-coverage effect persisted beyond the formal policy reversal and intensified during the 2023-2024 Medicaid unwinding. This is consistent with fear and administrative disruption compounding each other (Section 5.5). The persistence finding adds to the literature on restrictive policy changes generating lasting behavioral effects that outlive the policies themselves (Hacker, 2004; Bustamante et al., 2022; Bernstein et al., 2022).

Third, the California analysis supplies supporting evidence on the mechanisms of fear-based avoidance, gradient-based effects, and unique state results. Naturalized citizens experienced a 1.3 percentage point decline in Medi-Cal enrollment even though they were unambiguously exempt from the rule, and immigrant groups reported substantially greater documentation scrutiny when seeking coverage. Together, these results quantify the administrative environment in which chilling effects operate and show that fear spilled beyond the legally targeted population. Figures 1 and 2 summarize the legal scope of the rule, the populations it chilled, and how the two empirical arms fit together.

The remainder of the paper proceeds as follows. Section 2 provides background on the public charge doctrine and prior evidence on chilling effects, Section 3 describes data, and Section 4 presents methods. Section 5 reports results, with the national ACS child estimates first (Sections 5.1-5.6) and the supporting California adult evidence second (Sections 5.7-5.9). Section 6 discusses interpretation, policy implications, and limitations, and Section 7 concludes.

2. Background

2.1. What “public charge” is and how it has evolved

In U.S. immigration law, “public charge” is a ground of inadmissibility - a reason the federal government may deny a noncitizen entry to the country or, more commonly, deny their application to adjust status to lawful permanent resident (LPR, or “green card”). The legal test is forward-looking: an adjudicating officer at U.S. Citizenship and Immigration Services (USCIS) or Department of State consular assesses whether, in their judgment, the noncitizen is “likely to become a public charge”. How much the public charge doctrine matters in practice depends on three components: (i) what categories of benefit use count as evidence of likely dependence, (ii) which other personal characteristics (age, income, family ties, English proficiency) are weighed, and (iii) how heavily benefit use is weighted relative to those other factors. The 2019 rule finalized by the first Trump administration expanded components (i) and (iii), with Medicaid use being a new and more heavily weighted component. While how a public charge determination was made changed, it did not directly change the population eligible or affected: U.S. citizens were still not subject to public charge determinations, lawful permanent residents are at risk only at narrow re-entry margins and not for renewal of LPR status, and naturalized citizens are no longer subject to the doctrine in any form.

Why would a rule like this matter for Medicaid take-up by populations who are not at risk? The chilling-effects literature documents a recurring pattern: when restrictive immigration policy raises the perceived cost of using benefits, take-up declines among populations not formally subject to the policy. This is a behavioral effect rather than a legal one: families often cannot easily tell what does and does not count as “public charge use,” and families with noncitizen members or uncertain community ties may respond to the perceived threat by avoiding benefit interaction altogether. The economic logic is standard non-price participation cost (Currie, 2006): when the perceived cost of using a benefit rises, take-up falls, even when the formal eligibility rules are unchanged.

The paper’s main analytic population is U.S.-citizen children below 200 percent of FPL living in households with at least one noncitizen parent (“mixed-status families”). Federal Medicaid and CHIP rules treat these children identically to citizen children in citizen-only households. Their parents’ immigration status is not a factor in their own eligibility, parents are not required to disclose their own immigration status when applying on the child’s behalf in most states, and the federal “five-year bar” on most means-tested benefits for new lawful permanent residents (PRWORA section 403) does not apply to U.S.-citizen children regardless of parent status. The great majority of these children are eligible for Medicaid or CHIP if the household income test is met. The chilling-effect hypothesis is that entitled populations disenroll or fail to enroll because they fear that they or related family will be negatively affected due to public program use; in the case of this paper, parents perceive that interacting with the Medicaid agency on the child’s behalf may create immigration risk for themselves, or may have long-term negative consequences for their child.

The doctrine itself is old, with the concept of a charge dating to the colonial era. When communities sought to exclude settlers who might become dependent on local poor relief, Congress passed the Immigration Act of 1882, which barred admission to any person “likely to become a public charge.” For more than a century afterward, public charge determinations focused almost exclusively on whether an individual was likely to become primarily dependent on the government for cash subsistence. The standard was applied at the point of admission or, for immigrants already in the United States, at the point of adjustment to LPR status. In practice, public charge denials were fairly rare; the Immigration and Naturalization Service (INS) exercised broad discretion, and the universe of benefits considered was limited to cash welfare programs such as TANF, SSI, and state general assistance (Capps et al., 2018).

The modern regulatory framework took shape after the 1996 welfare reform legislation, a primary component of the Republican-led Congress’ Contract with America legislative agenda. PRWORA dramatically restructured immigrant eligibility for public benefits, creating a five-year waiting period for LPRs to access most federal means-tested programs and making undocumented immigrants categorically ineligible for nearly all federal benefits except emergency Medicaid. However, the 1996 law did not change the public charge standard itself - receipt of non-cash benefits was not negative factor in immigration determinations. In partial response to reported chilling effects PRWORA had already generated among immigrant communities (Kandula et al., 2004; Kaushal and Kaestner, 2005), the INS clarified the distinction in its 1999 Field Guidance, which explicitly stated that use of Medicaid (except long-term institutional care), SNAP, CHIP, and housing assistance would not be considered in public charge evaluations.

For nearly two decades after the 1999 Guidance, the public charge doctrine operated as a narrow and clearly defined concept: cash assistance and government-funded institutional care counted, while non-cash programs did not. While the distinct clarity between the two provided a bright-line rule that could be communicated to immigrant families, service providers, and enrollment workers, fear of government interaction has been persistent among immigrant communities.

2.2. The 2019 rule expansion

The Trump administration upended this framework beginning in 2017 when a leaked draft executive order in January 2017 signaled the administration's interest in expanding public charge determinations to include a wider range of public benefits. The Department of Homeland Security (DHS) placed the rule on the Unified Regulatory Agenda in October 2017, putting immigrant advocacy organizations on notice that formal rulemaking was coming. I note that these early signals were important because they started the informational cascade that could produce chilling effects months before any formal rule change.

The final rule specified that receipt of Medicaid (for non-pregnant adults under age 65), SNAP, or federal housing assistance for more than 12 months in any 36-month period would be a "heavily weighted negative factor" in public charge determinations under a "totality of the circumstances" test (84 Fed. Reg. 41,292).

A few features amplified the rule's chilling potential. First, it was widely misunderstood as applying well beyond its formal scope: surveys found that many immigrants (including naturalized citizens, refugees, asylees, and U.S. citizens in mixed-status families) believed that their own benefit use or family members' use could trigger public charge consequences (Bernstein, McTarnaghan, and Gonzalez, 2019; Gonzalez, Bernstein, and Karpman, 2020). Second, the rule explicitly excluded children's Medicaid and CHIP, but due to incomplete

information and the child-parent linked nature of these benefits, many families could not parse the distinction between program components that counted and those that did not. Third, even families that understood the scope faced a rational fear problem: the rule signaled that the government was interested in monitoring benefit use among immigrants, which raised the perceived cost of any interaction with enrollment systems.

Media coverage amplified the confusion. The September 2018 NPRM generated extensive coverage in both English- and Spanish-language media, frequently describing the rule in broad terms without fully conveying the exemptions. Community-based organizations worked to spread accurate information, but the rule's complexity, and uncertainty about its final form during the 14-month comment period, made clear guidance difficult. The result was an informational environment in which fear was both rational (the rule did apply to some immigrants and did involve Medicaid) and overinclusive (many families believed they were affected when they were not).

2.3. The litigation cycle

The rule's implementation was marked by extended litigation that added further uncertainty (Figure 3). In October 2019, federal courts in multiple circuits issued preliminary injunctions blocking implementation. In January 2020, the Supreme Court stayed most injunctions (the 7th Circuit exception meant Illinois remained enjoined), and the rule took effect in February 2020 in most states. New injunctions during the pandemic effectively halted implementation by late 2020, and the Biden administration formally restored the 1999 standard in March 2022. The second Trump administration released a final rule in July 2026 which restored much of the 2019 regulation and gave even more discretion to immigration officers.

The litigation cycle muddles interpretation because it creates a long period of uncertainty with no clear beginning and end to the main policy treatment. Even after injunctions, families could not be confident that the rule would not ultimately take effect - and even after the Biden reversal, the memory of the policy and possibility of its revival may have permanently altered immigrant behavior and kept some families from re-enrolling. As such, the behavioral response cannot be expected to track the formal legal status of the rule with precision.

2.4. The Medicaid unwinding

The Families First Coronavirus Response Act (March 2020) conditioned enhanced federal Medicaid funding on a continuous enrollment requirement: states could not disenroll beneficiaries during the public health emergency without giving up a significant amount of federal funding. This provision prevented routine eligibility redeterminations for more than three years. When it expired at the end of March 2023, states resumed redeterminations, and millions of beneficiaries were disenrolled - many for procedural reasons (failure to return paperwork, outdated addresses, administrative processing errors) rather than actual ineligibility (KFF, 2024). The unwinding is relevant because immigrant families may have been disproportionately vulnerable to procedural disenrollment: language barriers, unstable addresses, unfamiliarity with government correspondence, and fear of re-engaging with enrollment systems after years of avoidance all contribute to higher procedural churn among mixed-status families. Additionally, pre-existing chilling and the unwinding's administrative demands may compound each other - one motivation for the split-period decomposition in the empirical design (Section 5.5).

2.5. Prior evidence on chilling effects

Welfare reform precedent. The concept of chilling effects has deep roots in the immigration policy literature. The foundational evidence comes from the 1996 welfare reform, which

restricted immigrant eligibility for federal means-tested programs and created a five-year waiting period for new LPRs. Although PRWORA's formal provisions targeted specific groups of immigrants, its effects spilled over to populations that remained legally eligible: Kandula et al. (2004) provided early evidence that PRWORA reduced Medicaid enrollment among pre-1996 immigrants who were explicitly exempt from the law's restrictions. Using Current Population Survey data from 1994 to 2001, they estimated a 3 percentage point decline in Medicaid enrollment among eligible immigrants, compared with 1.6 percentage points for citizens. States that maintained universal immigrant coverage partially buffered the effect. Kaushal and Kaestner (2005) found even larger effects: uninsured rates among foreign-born unmarried women rose by 10-11 percentage points, and their children's uninsured rates rose by 13.5 percentage points. These effects could not be attributed to formal eligibility changes and were consistent with fear, confusion, and misinformation about who was affected.

The welfare reform episode established a pattern that would repeat with the public charge rule: a policy targeted at a subset of immigrants generated fear that radiated well beyond its formal scope - an informational mechanism, not a legal one (Section 2.1).

A useful comparison to chilling effects may be as the mirror image of the ACA's "welcome-mat" (or "woodwork") effect. When the ACA expanded Medicaid to low-income adults in 2014, coverage rose among the newly eligible and among people who had been eligible all along. Frean, Gruber, and Sommers (2017) show that a substantial share of the ACA's Medicaid enrollment gains came from previously eligible adults taking up coverage, and Hudson and Moriya (2017) document that the adult expansions produced measurable "welcome-mat" coverage gains among children whose own eligibility never changed - parents brought their already-eligible children through the door with them. In both directions, the operative margin is

take-up among the already-eligible, moved by policy signals about whether public coverage is welcoming or risky rather than by any change in formal eligibility. The public charge rule is the welcome-mat effect run in reverse: a signal that participation had become dangerous, transmitted through the same parent-mediated household channel, pushing eligible children out of coverage just as the expansion-era signal had pulled them in. This symmetry also matters for interpreting the paper's pre-period as early years sit inside the ACA's welcome-mat window (Section 5.2).

Immigration enforcement and safety-net participation. A parallel literature documents that immigration enforcement, independent of eligibility rules, can reduce safety-net participation. Watson (2014) provided seminal causal evidence using instrumental variables and variation in INS/ICE enforcement intensity across metropolitan areas. She showed that heightened enforcement reduced Medicaid participation among citizen children of noncitizens, and estimated that up to 75% of the decline in noncitizen Medicaid participation around welfare reform was explained by contemporaneous enforcement spikes rather than PRWORA itself. I note this as a nuance and potential weakness of this analysis: response to eligibility changes for immigrants is almost always concurrent with a heightened enforcement environment.

Alsan and Yang (2024) extended this research to the Secure Communities era. Using county-level rollout of Secure Communities between 2008 and 2013, they showed that the program's activation reduced SNAP and ACA enrollment among Hispanic citizen households. Effects were concentrated in mixed-status families, and the authors concluded that results were most consistent with network transmission of fear. Their estimate that ACA sign-ups would have been 22% higher absent Secure Communities shows how quantitatively important enforcement-driven chilling can be. Kravitz et al. (2024) linked county-level ICE detainer requests to lower Medicaid and SNAP enrollment, with effects stronger among households containing immigrant members.

Twersky (2022) showed that restrictive state immigration laws reduced Medicaid/CHIP enrollment among citizen children in immigrant families by up to 5.5 percentage points.

Collectively, these studies collectively establish that fear, uncertainty, and perceived government risk are sufficient to deter enrollment among populations with full legal entitlement to benefits, and that the public charge rule is an especially potent version of this dynamic as it explicitly links public benefit use to immigration consequences.

Evidence specific to the public charge rule. For the public charge episode itself, Barofsky et al. (2020) provided the strongest causal evidence to date. Using county-level administrative enrollment data for Medicaid, SNAP, and WIC from January 2015 through June 2019, they estimated a difference-in-differences model comparing counties with higher versus lower noncitizen population shares before and after the September 2018 announcement. They estimated approximately 260,000 fewer children enrolled in Medicaid, with the decline preceding any formal rule change, occurring among children explicitly exempt from the proposed rule, and concentrating in counties with the highest exposure.

My work extends beyond the Barofsky et al. design in two ways: First, the county-exposure approach treats noncitizen population share as the treatment intensity, which mixes true exposure with several confounded characteristics of high-noncitizen-share counties: more left-leaning state policy environments, denser sanctuary infrastructure, urban-rural composition, and contemporaneous coverage expansions. The individual-level mixed-status family proxy used here defines treatment at the household level, which is closer to the policy's actual mechanism (parental immigration exposure rather than residential neighborhood composition). Second, the Barofsky study ended in June 2019 - before the rule's formal implementation in February 2020, before the 2021 reversal, and well before the 2023-2024 Medicaid unwinding. The present paper

extends the evidence through the full policy cycle, including the unwinding period, which allows a direct test of whether chilling effects persist through reversal and grow under later administrative shocks.

While causal evidence of chilling effects and immigrant behavior is moderate, survey and descriptive studies are much more prevalent. KFF microsimulation estimated that 2.0 to 4.7 million Medicaid and CHIP enrollees could disenroll, including 600,000 to 1.8 million citizen children (Artiga, Garfield, and Damico, 2019). The Migration Policy Institute documented that participation in TANF, SNAP, and Medicaid declined twice as fast among noncitizens as among citizens during 2016-2019 (Capps, Fix, and Batalova, 2020). Urban Institute found that one in five adults in immigrant families reported avoiding benefits, including naturalized citizens and green card holders not subject to the rule (Bernstein, McTarnaghan, and Gonzalez, 2019).

Bernstein, Gonzalez, Karpman, and Zuckerman (2020) confirmed these patterns persisted into 2019. Bernstein, Gonzalez, Echave, and Guelespe (2022) showed that avoidance continued into 2021 after reversal, providing early evidence that the effects outlast the policy. Qualitative reports from health center staff confirmed that disenrollment and enrollment avoidance extended to lawfully present immigrants not subject to the rule (Tolbert, Artiga, and Pham, 2019).

The descriptive literature has also identified a key measurement constraint that the present paper addresses with restricted-use CHIS access. Most population surveys distinguish only between citizens and noncitizens, pooling lawful permanent residents (green card holders) with undocumented immigrants in the noncitizen category. Surveys have historically been unwilling to ask questions that directly identify undocumented status. Pooling these groups attenuates estimated chilling effects (Section 3.2). Restricted CHIS data (via the UCLA Data Access Center) provides a four-level variable that separates green-card holders from noncitizens without

green cards, which allows a substantively different decomposition of California adult chilling effects than has been previously possible.

Evidence on related healthcare utilization is more limited but growing. Haro-Ramos et al. (2026) found that the 2018 announcement was associated with a 2.1 percentage point increase in self-pay ED visits among likely undocumented patients in LA County, while Choi et al. (2023) showed that uninsured immigrants significantly delayed prenatal care initiation after the rule changes. The utilization findings reinforce the coverage findings: families avoiding Medicaid enrollment under the public charge rule still required medical care, but increasingly accessed it through self-pay or delayed channels rather than through the formal public coverage they were eligible for. The fear-based avoidance gradient identified in this paper allows direct measurement of the household-level decision that produces this pattern (Section 5.8).

Administrative burden as a mechanism. The administrative burden framework (Herd and Moynihan, 2018) provides theoretical grounding for how the public charge rule could reduce enrollment without changing eligibility by distinguishing learning costs, compliance costs, and psychological costs of interacting with government programs. The public charge rule raised all three at once: it created a complex regulatory framework that even attorneys had difficulty parsing (learning costs), signaled that enrollment systems would scrutinize immigrant applicants more carefully (compliance costs), and associated benefit use with immigration risk (psychological costs). These burdens fall disproportionately on low-income, limited-English-proficiency families (Moynihan, Herd, and Harvey, 2015; Heinrich, 2016). Mixed-status families face an additional layer of burden because the enrollment process for citizen children often requires information about parents' income and household composition, which creates opportunities for questions that immigrant parents find threatening.

A growing empirical literature documents the magnitudes of these channels. Bhargava and Manoli (2015) showed that simplified communications and pre-completed forms can substantially raise take-up of EITC and other safety-net programs even among populations with no formal eligibility change. Finkelstein and Notowidigdo (2019) extended this finding to SNAP, showing that information and assistance interventions raise enrollment by roughly 25%. The implication of these studies is that the marginal cost of “one more administrative step” is large for low-income participants whose time and attention are already stretched thin.

Within this framework, the 2019 rule is best understood as a negative information shock about the safety of program participation. The interventions studied by Bhargava and Manoli (2015) and Finkelstein and Notowidigdo (2019) raised take-up by delivering positive information: you are eligible, the process is simple, participation is safe. The public charge announcement delivered the opposite message through national and community media - participation is monitored, benefit records can be held against your family, and the boundary between “counted” and “exempt” programs is too complicated to verify. For a family that cannot cheaply confirm its own exemption, the rational response to that signal is to withdraw from the enrollment system entirely. This aligns with the psychological-cost channel Herd and Moynihan (2018) identify, activated at population scale by a single regulatory announcement: no formal rule needs to take effect for coverage to fall and the information shock itself is the treatment. The timing of the observed enrollment response, which begins with the September 2018 announcement well before the rule’s February 2020 implementation, is consistent with this interpretation. In practice, the rule added several psychological steps: enrolling now signals immigration risk, enrolling means generating government records that could later be used adversely, and enrolling triggers

questions that the family fears it cannot safely answer – with each added step plausibly worth several percentage points of enrollment among the marginal eligible household.

The documentation-demand evidence developed in this paper measures one of these psychological channels directly and shows it to be a durable feature of the enrollment environment, not a short-lived artifact of the 2019-2021 policy window (Section 5.9).

Policy feedback and persistence. This paper also attempts to connect the public charge episode to the political science literature on policy feedback. Hacker (2004) developed the concept of “policy drift” to describe how the welfare state can retrench not through overt program cuts but through the failure to update policies in the face of changing circumstances. Hacker and Pierson (2019) extended this framework to argue that policy changes can generate self-reinforcing feedback loops. In the immigration context, the public charge rule created a feedback dynamic in which fear reduced enrollment, which reduced the base of beneficiaries who might advocate for inclusive policies, which in turn made it easier for future administrations to consider similar restrictions.

The persistence hypothesis, that effects continue after formal reversal, follows directly from this framework: Once families disenroll from Medicaid, re-enrollment requires navigating the same administrative system that generated fear in the first place. The sunk costs of learning how to avoid the system may exceed the costs of re-engaging, and if families perceive that the policy environment may shift again (an entirely rational perception given recent political cycles), the option value of staying out increases. The result is a ratchet effect: it is easier to scare families away from programs than to bring them back. Bustamante et al. (2022) provided empirical evidence, estimating that 108,000 to 193,000 Latino immigrants in California continued to avoid Medi-Cal even after the Biden administration reversed the rule. This paper tests the hypothesis

more formally by examining whether the gap narrowed after the 2021-2022 reversal and whether it widened during the 2023-2024 unwinding.

3. Data

3.1. National analysis: American Community Survey

Data source and access. The national analysis uses ACS microdata accessed through IPUMS USA (Ruggles et al., 2024) for survey years 2014 through 2024. The ACS is the largest household survey in the United States outside the decennial census, sampling approximately 3.5 million addresses per year. It provides nationally representative individual-level data on demographics, household composition, income, health insurance coverage, and citizenship status. Survey responses are collected on a rolling monthly basis throughout the year, with each annual release representing a full 12 months of data collection.

The ACS identifies whether each individual is a U.S. citizen by birth, citizen by naturalization, or noncitizen. It does not distinguish among noncitizen types - lawful permanent residents, temporary visa holders, and undocumented immigrants. This limitation is common to all major federal surveys, and it means the mixed-status family definition necessarily pools families with different types of noncitizen parents. The pooling likely attenuates the estimates relative to the effect among the most legally vulnerable families, because families with undocumented members have the most to fear from public charge-style policies. A family with an undocumented parent faces qualitatively different risks than one with a parent holding a green card near naturalization.

Sample construction. The preferred sample includes U.S. citizen children ages 0 to 18 in households below 200% of FPL. This income restriction is substantively important for two

reasons. First, the chilling-effect hypothesis predicts the largest enrollment response among children plausibly eligible for Medicaid or CHIP. Children in families well above the income threshold have little reason to participate in public coverage programs and therefore little reason to respond to changes in the perceived cost of enrollment; including them dilutes the estimated effect without adding information about the relevant behavioral margin. Second, the below-200% FPL threshold approximates the upper bound of Medicaid/CHIP eligibility in most states. Exact thresholds vary by state, child age, and program (some states extend CHIP eligibility to 300% FPL or higher), but the 200% FPL cutoff captures the overwhelming majority of the population for whom public coverage is a realistic option.

The full citizen-child sample contains 6,482,027 observations across eleven survey years (2014-2024). The preferred below-200% FPL sample contains 2,077,869 observations: 389,899 (18.8%) in mixed-status families and 1,687,970 (81.2%) in citizen-only families. The two added pre-announcement years contribute 615,565 children in 2014 and 610,140 in 2015, which extends the pre-period available for parallel-trends testing from two survey years to four. I present full-sample estimates for all income levels ($N = 6,482,027$) as secondary specifications (Section 5.3). Because 2014 and 2015 fall inside the ACA's coverage-expansion rollout, Section 5.2 discusses how that overlap affects the interpretation of the extended pre-period leads.

Treatment group definition. I classify children as living in mixed-status families if at least one identified parent in the household roster is a noncitizen, while citizen-only families are those in which all identified parents are U.S. citizens. The child must be a U.S. citizen in both groups. Children whose parents cannot be identified through the IPUMS household-roster pointers (~4% of the eligible-age citizen-child sample) are excluded. This share is small and approximately stable across calendar time within the analytic window, so the exclusion is unlikely to drive the

post-2018 differential trend that the DiD identifies. If anything, families in which household relationships are difficult to enumerate are more likely to be in mixed-status households, so excluding them is likely to mildly attenuate the headline estimate.

This classification captures the household-level immigration exposure central to the chilling-effect mechanism. The theory does not predict that children themselves respond to the public charge rule - children do not make their own enrollment decisions – but rather that parent-level exposure in mixed-status families, who perceive a link between benefit use and immigration risk, alter their children’s enrollment behavior.

Outcome variables. The primary outcome is current Medicaid enrollment. Secondary outcomes include any public coverage, any health insurance coverage, and uninsurance. The distinction between Medicaid and any public coverage matters because some children may shift between programs (for example, Medicaid to CHIP) in response to the public charge environment. The uninsurance measure captures the most consequential outcome (children with no coverage at all) but may be noisier if some adjustment involves shifts to employer-sponsored or individual-market insurance.

Controls and summary statistics. All ACS models use person-level survey weights and include state and year fixed effects, which absorb time-invariant state characteristics and national trends common to all states. Individual-level controls include child age (entered as dummy variables), sex, and race/ethnicity (non-Hispanic White, non-Hispanic Black, Hispanic, Asian/Pacific Islander, and Other). Standard errors are clustered at the state level to account for within-state correlation in outcomes over time and across individuals.

Table 1 presents pre-period (2014-2018) weighted means. Children in mixed-status families are modestly younger (8.24 vs. 9.04), equally likely to be female, and substantially more likely to be

below 200% FPL (0.584 vs. 0.335). Pre-period Medicaid enrollment is higher (0.602 vs. 0.337) and pre-period uninsurance is also higher (0.071 vs. 0.037), reflecting lower income and baseline chilling effects that predate the 2019 rule. Despite these level differences, the identifying assumption requires parallel trends, not identical levels.

3.2. California analysis: California Health Interview Survey

What CHIS is and how it is used. The California Health Interview Survey (CHIS) is the largest state health survey in the United States, administered annually by the UCLA Center for Health Policy Research. It samples approximately 20,000-25,000 adults per cycle, is conducted in English, Spanish, Chinese (Mandarin and Cantonese), Korean, Vietnamese, and Tagalog, and is one of the only population-based surveys in the United States that separately identifies U.S.-born citizens, naturalized citizens, and noncitizens. Use of CHIS is also intended to mitigate the well-known non-response bias that English-only surveys often have.

The California analysis uses two CHIS data products, the public-use file and the restricted-use adult file, and which file generated which result depends on which variables a given specification requires; I make the file-by-result mapping explicit so the reader can trace every California estimate to its source (Table 2).

The following specifications are estimated on the public-use adult file:

- The three-group multi-group DiD comparing U.S.-born citizens, naturalized citizens, and pooled noncitizens (using public-use CITIZEN2) on the four standard outcomes: Medi-Cal enrollment, ED visit in past 12 months, having a usual source of care, and forgone care. The naturalized x Post coefficient on Medi-Cal reported as supporting evidence in Section 5.8 is a public-use estimate.

- The documentation-demand specification in Section 5.9, which uses the public-use binary indicator for whether the respondent was asked for a Social Security number or citizenship documentation when seeking health coverage. The item is in the public-use adult file; universe, timing, and reference-group choices are described in Section 4.2.
- The fear-based benefit-avoidance gradient (the public-use AL99-equivalent indicator on whether the respondent ever decided not to apply for a non-cash benefit because of immigration concerns), used in Section 5.8 alongside the restricted-use AL104 12-month follow-up. The public-use indicator is the lifetime-recall version; the 12-month follow-up is restricted only.
- All summary statistics in Appendix Exhibit A1 and the poverty-stratified heterogeneity analyses reported in the appendix.

The following specifications require restricted-use variables:

- The four-level CITIZEN1 decomposition, and any analysis that separates undocumented adults from green-card holders. This includes the age-stratified four-level DiD (Section 5.7), the two-window decomposition that separates federal chilling from the California full-scope Medi-Cal expansion calendar, and the green-card-versus-naturalized fear-avoidance contrast.
- The 12-month-recall fear-avoidance variant (AL104); only the lifetime-recall version (AL99) is on the public-use file.
- The county-enforcement heterogeneity test (Appendix Exhibit A18), which requires the full county FIPS code (FIPS_CNT) suppressed in the public-use file.

Sample composition and the four-level citizenship structure. Across both CHIS files, the pooled 2015-2024 adult sample (N = 220,925) contains 168,525 U.S.-born citizens (76.3%),

35,656 naturalized citizens (16.1%), and 16,744 noncitizens (7.6%). The public-use CITIZEN2 variable pools the last group into a single “noncitizen” category. The restricted-use CITIZEN1 variable splits that category into noncitizens with a green card (lawful permanent residents) and noncitizens without a green card. The literature treats the no-green-card group as the closest available proxy for undocumented status, recognizing that some respondents in this category hold temporary visas, DACA, TPS, or other non-LPR documented status.

The four-level structure matters substantively because public charge matters mainly for adjustment-of-status applications. Lawful permanent residents have already cleared adjustment and face the rule only at narrow re-entry margins; undocumented adults face it at any future adjustment attempt. Pooling the two in CITIZEN2 mechanically attenuates estimated chilling effects on the most directly exposed population. The four-level split also makes visible the near-exact overlap between California’s contemporaneous full-scope Medi-Cal expansions and the federal chilling window - the reason the within-state adult coverage estimate is not a clean chilling estimate (Section 5.7).

Naturalized citizens are valuable to the design for the opposite reason: unambiguously exempt from public charge yet embedded in immigrant households, social networks, and language-minority media environments - so any enrollment decline among them evidences network-mediated fear rather than direct legal exposure (Section 5.8).

I restrict the CHIS adult sample to below 200% FPL but also run full-FPL analyses as its outcomes are not limited to the lower-income tranches; however, poverty remains important for interpretation, so I control for it in all models and report poverty heterogeneity in the appendix. Descriptive statistics are in Appendix Exhibit A1.

Outcomes. CHIS contributes four standard outcomes available in both files: Medi-Cal enrollment, ED visit in the past 12 months, having a usual source of care, and forgone care in the past 12 months. Two additional restricted-only outcomes are central to this paper: the fear-based benefit-avoidance items AL99 and AL104 (Section 5.8), and the documentation-demand item AL19V2 (Sections 4.2, 5.9).

Restricted-use access process. The restricted-use CHIS adult file is accessed through the UCLA DAC. Researchers submit a vetted project application and then submit Stata or SAS programs that DAC staff execute against the confidential file in a secure environment, returning disclosure-reviewed output with small-cell suppression rules applied. All restricted-use results in this paper were produced through this process (all specifications listed above). Because the survey-weighted estimates and their replicate-weight variances are returned directly, each restricted-use figure reported here is traceable to a specific disclosure-reviewed output cell.

Variance estimation. The CHIS files do not include the sampling design variables needed for Taylor-series linearization variance estimation and instead include 80 balanced repeated replication (BRR) replicate weights and/or jackknife replicate weights. I use adult survey weights in all CHIS estimates, report heteroskedasticity-robust standard errors as the primary inference, and treat BRR/jackknife replicate-weight estimates as a survey-design sensitivity analysis (see appendix). I do not rely on the smaller replicate-weight standard errors to strengthen claims that are weak under the more conservative primary inference.

4. Methods

4.1. ACS identification strategy

Difference-in-differences framework. The primary specification is:

$$Y_{ist} = \alpha + \beta (\text{MixedStatus}_i \times \text{Post}_t) + \gamma \text{MixedStatus}_i + \delta_s + \theta_t + X_i \lambda + \varepsilon_{ist}$$

where Y_{ist} = the Medicaid enrollment indicator for child i in state s in year t , MixedStatus_i = an indicator for living in a mixed-status family, Post_t = an indicator for survey years 2019 and later (reflecting the September 2018 announcement), δ_s and θ_t = state and year fixed effects, X_i = individual controls (age dummies, sex, race/ethnicity), and ε_{ist} represents an error term. Citizen children in mixed-status families are considered as the “treated” group, while children in citizen-only families serve as “untreated” controls. The coefficient of interest is β , which captures the differential change in Medicaid enrollment for children in mixed-status families relative to citizen-only families after the announcement. Standard errors are clustered at the state level, the level at which Medicaid policy is administered. The preferred sample is below 200% FPL. Full-sample results are secondary (Section 3.1). Figure 4 summarizes the exposure framework behind this design.

Event-study and split-period specifications. I estimate event-study models interacting mixed-status family status with year indicators. The coefficients for years before 2018 (the leads) test for pre-policy divergence - if the parallel trends assumption holds, these should be close to zero and jointly insignificant – while the coefficients for 2019 onward trace the dynamic treatment path. I conduct a joint Wald test of the pre-period coefficients using the estimated variance-covariance matrix, which is preferable to examining individual leads in isolation because it accounts for the joint distribution. The analysis window provides four pre-treatment years (2014-2017), with the two earliest years overlapping the ACA’s coverage-expansion rollout, which is itself a shock to take-up among the already-eligible (Section 2.5). Section 5.2 therefore reports the individual leads, the joint test, and the sensitivity of the pre-trend evidence across specifications rather than relying on a single summary p-value. A split-period specification then

separates pre-unwinding (2019-2022) from unwinding (2023-2024) effects by replacing the single post indicator with two period indicators, each interacted with mixed-status. This decomposition is important for two reasons: First, the continuous enrollment requirement during the pandemic limited disenrollment of existing beneficiaries (Section 2.4), which may have mechanically attenuated the estimated chilling effect during that sub-period. Second, the unwinding created new administrative barriers that may have disproportionately affected immigrant families.

Secondary specifications. Several secondary specifications test the robustness and interpretation of the main ACS results. A placebo test restricts the sample to citizen-only families and compares those in states with high noncitizen population shares to those in low-noncitizen-share states: if the mixed-status effect reflects a genuine chilling response specific to families with noncitizen members, this placebo should yield null results. I also report full-sample ACS results, an alternative timing specification using the February 2020 implementation date rather than the September 2018 announcement, earlier-treatment sensitivity analyses that treat the policy threat as beginning during the 2017-2018 regulatory buildup, and a separate state-by-year fixed-effects event study that absorbs any state-specific shocks shared by mixed-status and citizen-only children within a given year. Because these exercises are secondary to the below-200% FPL child Medicaid design, they are reported mainly in the appendix.

4.2. CHIS identification strategy

Multi-group difference-in-differences. The preferred CHIS specification estimates separate post-2018 coefficients for noncitizens and naturalized citizens relative to U.S.-born citizens:

$$Y_{it} = \alpha + \beta_1 (\text{Noncitizen}_i \times \text{Post}_t) + \beta_2 (\text{Naturalized}_i \times \text{Post}_t) + \gamma_1 \text{Noncitizen}_i + \gamma_2 \text{Naturalized}_i + \theta_t + X_i' \lambda + \varepsilon_{it}$$

where Y_{it} = the outcome for adult i in survey year t , Noncitizen_i and Naturalized_i = citizenship-group indicators (U.S.-born citizens omitted), Post_t = an indicator for survey years 2019 and later, θ_t = survey-year fixed effects, X_i = controls for age, sex, race/ethnicity, and poverty, and ε_{it} represents an error term. The multi-group specification is more informative than a two-group comparison because it separately identifies effects for populations with very different policy exposure. Noncitizens include those most directly affected by the rule, though the public-use CHIS pools them with green card holders for whom risk was lower. Naturalized citizens were fully exempt but may have responded through network-mediated fear.

Event studies and documentation-demand analysis. CHIS event studies interact group indicators with year dummies, providing three pre-treatment years (2015-2017) and six post-treatment periods (2019-2024). Joint Wald tests of pre-period coefficients are conducted for each outcome. As an additional diagnostic, I also estimate pairwise event studies comparing noncitizens with U.S.-born citizens and naturalized citizens with U.S.-born citizens, and I check those samples for missingness, empty year-by-group cells, and event-study saturation.

For the documentation-demand measure, I estimate cross-sectional post-2019 models on the immigrant-only respondent universe, which is the CHIS sampling universe for the AL19V2 item. Naturalized citizens are the reference category; noncitizens (pooled green-card holders and adults without green cards in the public-use file) are the contrast. Restricting to the immigrant universe avoids the spurious U.S.-born baseline of structural zeros and aligns the regression with the question's intended sampling frame. Because the item is unavailable before 2019, this is a within-immigrant cross-sectional gradient: it measures the size of the administrative vulnerability gap during the relevant policy period, not whether the public charge rule changed documentation practices.

Secondary CHIS specifications include two-group models comparing pooled noncitizens with U.S.-born citizens. Other alternative specifications include COVID sensitivity checks by dropping the 2020-2021 survey years and adding a COVID interaction term, heterogeneity analyses stratified by poverty level, and alternative treatment timing using 2020 (implementation) rather than 2019 (announcement) as the post-period threshold.

County-enforcement heterogeneity. To estimate whether there was significant variation in outcomes where enforcement may have been significantly different (i.e. sanctuary cities), I create a county-enforcement heterogeneity specification that interacts the four-level CITIZEN1 x Post DiD with a binary high-enforcement county indicator merged on county FIPS code by survey year. For each of the four standard outcomes, the model includes group x Post interactions for naturalized citizens, green-card holders, and no-green-card adults (U.S.-born citizens omitted), plus triple interactions of each group x Post term with the high-enforcement indicator, controlling for age, sex, race, below-200% FPL, survey-year fixed effects, and CITIZEN1 fixed effects. The group x Post coefficients identify the pooled response in the low-enforcement (San Francisco area of responsibility, or AOR) counties; the triple-interaction coefficients identify the differential response in the high-enforcement (Los Angeles + San Diego AOR) counties. Variance is estimated via 80 CHIS replicate weights using the design-supplied jackknife multiplier. The specification is run on the full restricted-use adult sample for 2015-2024 (N = 220,925), and all observations match a county-year. Two features limit what this test can identify: (1) the underlying enforcement values are AOR-level rather than within-AOR county-level, so the triple-interaction coefficients should be read as Southern-versus-Northern California enforcement-intensity contrasts, not within-AOR county gradients, and (2) because 2015-2021 enforcement values are back-filled with the AOR's 2022 value, the high-enforcement

indicator is essentially time-invariant within the AOR-pair partition during the federal chilling window, so the test is a between-AOR contrast on the level of the post-period response rather than a time-varying enforcement-shock design. The specification is reported in the appendix.

5. Results

5.1. Raw outcome trends (survey-weighted)

I present raw data before regression estimates as a contextual and framing exercise. Figure 5 presents weighted mean Medicaid enrollment rates over calendar time for citizen children in mixed-status versus citizen-only families in the below-200% FPL sample. The two groups track closely during the pre-announcement period (2014-2018), consistent with the parallel trends assumption. After the September 2018 announcement, a visible gap opens and persists through 2024, with mixed-status children declining relative to citizen-only children: the gap appears to widen further in 2023-2024, coinciding with the Medicaid unwinding. Appendix Figure A12 extends this to a 2x2 panel showing Medicaid and uninsured outcomes for both samples, confirming that the divergence is concentrated in the low-income sample.

5.2. Event study results (non-causal)

Figure 6 presents event-study coefficients for Medicaid enrollment in the below-200% FPL sample (2018 omitted), estimated on the full 2014-2024 window with four pre-announcement leads. No individual lead is statistically significant at the 5% level: 2014 (+0.9 pp, 95% CI -0.8 to +2.6, $p = 0.29$), 2015 (+1.5 pp, 95% CI -0.2 to +3.1, $p = 0.096$), 2016 (+0.4 pp, $p = 0.70$), and 2017 (+0.2 pp, $p = 0.76$). The joint Wald test of all four leads yields $p = 0.14$, so the preferred specification does not reject parallel pre-trends.

I qualify these results in a number of ways: First, extending the pre-period weakens the joint pre-trend evidence relative to the two-lead window: the joint p-value falls from 0.92 with 2016-2017 leads alone to 0.14 with all four leads, the 2014 and 2015 leads are positive, and the lead profile drifts gently downward into the reference year. The two leads immediately preceding the announcement (2016 and 2017) are tightly near zero, and the earliest leads coincide with the ACA's 2014-2015 coverage-expansion rollout - a welcome-mat period (Section 2.5) in which take-up among already-eligible, lower-baseline-coverage mixed-status families plausibly rose slightly faster than among citizen-only families. However, it could be interpreted that the 2014-2015 leads are evidence of a slow differential decline that predates the rule – I find modest support for that reading and the design cannot definitively exclude it. Second, the pre-trend evidence is weaker in two secondary specifications: the state-by-year fixed-effects event study is borderline and the DROP-7 sample rejects (Sections 5.4, 5.6). Third, the full-sample uninsured event study rejects its joint pre-trend test decisively ($p < 0.001$). As such, I frame Medicaid enrollment in the below-200% FPL sample as the main outcome-sample combination.

Two features of the estimates reduce concerns of these qualifications and strengthen the headline result: (1) the residual lead drift is small relative to the treatment-period response - the largest lead (2015, +1.5 pp) is less than half the size of the 2020 coefficient, and the post-period break is sharp rather than a continuation of a smooth trend, and (2) mixed-status children entered the announcement period with, if anything, slightly elevated relative enrollment, so the post-2018 gap represents a reversal of sign, not an extrapolation of the pre-period path.

The post-period path more fully fleshes out the dynamics of the chilling effect: the 2019 coefficient is -0.9 pp ($p = 0.12$), suggesting the effect was emerging but not yet statistically distinguishable from zero in the first year after the announcement. This makes sense because the

announcement came in September 2018, so the 2019 ACS captures a mix of pre- and post-announcement months. The effect deepens sharply in 2020 to -2.5 pp ($p = 0.010$), remains significantly negative in 2021 at -1.6 pp ($p = 0.013$), softens in 2022 to -0.9 pp ($p = 0.142$), and then re-intensifies during the Medicaid unwinding: -1.9 pp in 2023 ($p = 0.001$) and -3.3 pp in 2024 ($p < 0.001$). This dynamic pattern is inconsistent with a short-lived announcement shock that dissipated once the rule was vacated.

5.3. National child coverage results: main estimates

Table 3 reports the core ACS estimates. Among citizen children below 200% FPL, those in mixed-status families saw a **2.0** percentage point decline in Medicaid enrollment ($SE = 0.005$, $p < 0.001$; $N = 2,077,869$). Declines were nearly identical for any public coverage (-2.0 pp, $p < 0.001$), indicating no meaningful offset from other public programs. The uninsured estimate is positive but imprecise (+0.8 pp, $p = 0.115$).

In context of magnitude, a 2.0 percentage point decline from a baseline Medicaid enrollment rate of approximately 60% is a 3.3% relative decline. A 95% confidence interval (-3.1 to -1.0 pp) implies approximately ~50,000 fewer enrolled children - smaller than Barofsky et al.'s (2020) county-level estimate of approximately 260,000, though the designs are not directly comparable (Section 6.2).

Two features of the headline estimate deserve closer attention: First, the gap between the Medicaid coefficient (-2.0 pp, highly significant) and the uninsured coefficient (+0.8 pp, $p = 0.115$) implies that roughly 40% of the lost Medicaid enrollment moves into the uninsured category and roughly 60% moves into other coverage - most likely subsidized Marketplace plans, employer-sponsored coverage shifts within the household, or CHIP-reporting margins not separately captured in the ACS public-use coding. The directional interpretation is that families

experiencing chilling-induced Medicaid loss are partially but not fully buffered by alternative coverage. The Medicaid coefficient is therefore the appropriate measure of the chilling effect on the public-coverage margin, while the uninsured coefficient is best read as a lower bound on the population-level insurance disruption. Second, both endpoints of the 95% confidence interval are economically meaningful and large relative to the year-to-year movement in Medicaid enrollment among the comparison group during the pre-announcement window.

The full-sample (all-income-levels) specification attenuates the Medicaid coefficient to -0.5 pp ($p = 0.35$; $N = 6,482,027$). The attenuation is consistent with dilution from higher-income families for whom Medicaid is not the relevant coverage margin. Restricting to the eligible population sharpens the test rather than narrowing it to a non-representative subgroup. I report the full-sample results in Appendix Exhibit A2 for completeness and as a calibration check.

Income-band sensitivity. The below-200% FPL threshold is a blunt approximation of state Medicaid/CHIP eligibility, which varies by state, child age, and program component. I therefore re-estimate the main DiD on three additional nested income bands. Across <300% FPL (-2.47 pp, $p < 0.001$, $N = 3,194,465$), <200% FPL (-2.16 pp, $p < 0.001$, $N = 2,077,869$; primary), <138% FPL (-1.81 pp, $p = 0.003$, $N = 1,312,969$), and <100% FPL (-2.07 pp, $p = 0.002$, $N = 850,857$), the headline coefficient is stable in magnitude and remains statistically significant in every band. I interpret this as the estimate not being driven by children just below the 200% threshold, where eligibility uncertainty is highest - narrowing to the deepest-poverty band, where eligibility is essentially universal under any state's rules, yields the same qualitative result.

Decomposing the control group. A natural concern is that the citizen-only ACS control group itself contains immigrant-connected families (specifically, families with a naturalized parent) for whom chilling effects may operate even though no one in the household is a noncitizen.

Approximately 10.1% of the original below-200% FPL citizen-only control group has at least one naturalized parent. To test whether this contaminates the headline estimate, I decompose the citizen-only control group into U.S.-born-parent families (no naturalized parent in the household) and naturalized-parent families (at least one naturalized parent, no noncitizen parent) and re-estimate the DiD on the three-way comparison. Relative to the U.S.-born-parent baseline, citizen children in mixed-status families experienced a 1.82 percentage point Medicaid decline post-announcement (SE = 0.006, $p = 0.003$); naturalized-parent families experienced a directionally opposite +1.41 pp change (SE = 0.007, $p = 0.064$); and the immigrant-connected pairwise contrast, mixed-status versus naturalized-parent, is sharply negative at -3.28 percentage points (SE = 0.007, $p < 0.001$). The three-way decomposition therefore strengthens the paper's identification: the chilling effect on citizen children's Medicaid coverage concentrates in households where a parent faces direct immigration risk rather than across immigrant-connected households broadly (Figure 7). Over the extended 2014-2024 window, neither arm's event study rejects its joint pre-trend test (mixed-status $p = 0.108$; naturalized-parent $p = 0.140$), though both joint p -values reflect the same gently positive early leads discussed in Section 5.2. I treat the naturalized-parent $\times$ Post coefficient as supporting evidence and note that the mixed-vs-naturalized-parent contrast does not depend on the naturalized-parent pre-trend assumption precisely holding.

5.4. State policy environment heterogeneity

The federal public charge rule landed on fifty-one very different state Medicaid policy environments, and a natural question with direct policy stakes is whether inclusive state environments buffered the chilling effect. Seven jurisdictions extended Medicaid (or near-equivalent state-funded coverage) to undocumented adults during the 2018-2024 paper window:

California (three-phase, 2020-2024), Illinois (65+ in December 2020, 55-64 in May 2022, 42-54 in July 2022), Oregon (Healthier Oregon, July 2022), New York (pregnant April 2022, 65+ January 2024), Colorado (OmniSalud and pregnant), Washington (Apple Health Expansion, July 2024), and the District of Columbia (long-standing). These expansions are proxies for a broader immigrant-supportive state Medicaid policy environment (outreach intensity, sanctuary signaling, navigator infrastructure, and parental enrollment experience that translates to citizen children's coverage) rather than a direct mechanism of federal public charge.

Table 4 reports two complementary specifications. First, declines were sharper outside the seven aforementioned states with inclusive jurisdictions: excluding the seven (the “DROP-7” sample, $N = 1,541,715$) tightens the chilling estimate to -2.59 percentage points on Medicaid ($SE = 0.007$, $p = 0.0003$), which is about 30% larger in magnitude than the pooled headline figure (Figure 8). Second, the corresponding national triple-DiD keeps all states and interacts the chilling term with an expansion-state indicator: the non-expansion-state baseline effect is -2.61 pp ($SE = 0.007$, $p < 0.001$), and the expansion-state extra coefficient is +0.79 pp - positive, implying a smaller chilling effect in inclusive states, but imprecise and not statistically distinguishable from zero ($p = 0.34$).

The substantive interpretation is that chilling is sharper where the state environment is not inclusive, and that states with more immigrant-inclusive appear to partially and significantly offset the federal rule. The point estimates move in the buffering direction in both specifications, but the formal interaction test is underpowered to declare the difference significant. I also provide two caveats to discipline this interpretation: First, the DROP-7 event study inherits the extended pre-period issue from Section 5.2 in amplified form: its 2014 lead is +2.1 pp ($p = 0.015$) and its joint pre-trend test rejects ($p = 0.019$). The DROP-7 point estimate should

therefore be read as a robustness/heterogeneity result, not a co-primary specification. Secondly, the expansion indicator is a coarse proxy for a bundle of state policies (coverage generosity, outreach, sanctuary signaling, navigator infrastructure) that this design cannot separate; identifying which component does the buffering requires finer state-policy variation - a potential target for future extension work. At a minimum, this pattern rules out the opposite hypothesis that inclusive state environments concentrate chilling, and suggests the main estimate is not wholly driven by the inclusive states.

5.5. Unwinding decomposition

The split-period specification formalizes the event-study pattern, with the pre-unwinding Medicaid estimate (2019-2022) being -1.7 pp ($p = 0.001$); the unwinding-period estimate (2023-2024) deepens to -2.8 pp ($p < 0.001$). This difference is consistent with a compounding dynamic, though the differences are not large enough to confirm a significant difference between the two. I argue deepening should not be read as a pure continuation of the original chilling effect: while unwinding created new administrative demands (renewal forms, redetermination letters, and eligibility verification) that affected all beneficiaries, families that had already disengaged during the public charge period were likely more vulnerable: they may not have received renewal notices, may not have responded to requests for updated information, or may have actively avoided re-engaging with a system they associated with immigration risk. Full-sample results are attenuated and insignificant (Appendix Exhibit A10). The below-200% FPL uninsured result shows a marginally significant unwinding-period effect (+1.0 pp, $p = 0.070$), directionally consistent with the Medicaid findings.

The 1.7 pp pre-unwinding estimate is notable on its own: the 2019-2022 window spans the entire policy cycle (Sections 2.2-2.4), including the continuous-enrollment protection period (Section

2.4). That a 1.7 pp Medicaid gap opens and persists across this entire span, including the protective continuous-enrollment window, implies that the chilling channel must operate primarily through foregone new enrollment rather than through active disenrollment of existing beneficiaries. The pre-unwinding coefficient therefore identifies a clean foregone-enrollment channel that later administrative shocks (the unwinding) compounded but did not start.

The fact that the effect did not fully attenuate during the post-reversal portion of the pre-unwinding window (2022) is also suggestive of prolonged effects. The effect did not collapse to zero after the March 2022 reversal. This is consistent with the policy-feedback hypothesis in Section 2.5. The pre-unwinding period therefore contains both an in-period chilling effect (2019-2021) and a post-reversal residual effect (2022) that the population-survey design cannot separately decompose, but that the persistence of the gap implies is real.

5.6. Secondary ACS checks

The secondary ACS exercises mainly help calibrate the preferred design rather than replace it.

The full-sample attenuation is discussed in Section 5.3.

The placebo tests provide more useful falsification evidence. Restricting the sample to citizen-only families and comparing those in high- versus low-noncitizen-share states yields null estimates for all three outcomes: Medicaid (-0.4 pp, $p = 0.41$), uninsured (+0.2 pp, $p = 0.37$), and any coverage (-0.2 pp, $p = 0.37$). These nulls confirm that the main estimate is specific to mixed-status families rather than a generic trend affecting all families in states with large immigrant populations.

The triple-difference and earlier-treatment specifications are reported in the appendix. The triple-difference high-noncitizen-share interaction is null (Appendix Exhibit A15), and the earlier-treatment exercise is directionally consistent with the policy threat building gradually, but it is

not central to identification. A separate state-by-year fixed-effects event study yields the same qualitative pattern as the preferred ACS design, with an average post-announcement coefficient of -1.9 percentage points ($p < 0.001$); its joint pre-trend p-value over the extended window is borderline ($p = 0.078$), driven by the same positive 2014 lead discussed in Section 5.2 (Appendix Exhibits A16-A17). I retain these exercises as supporting calibration checks, not headline results.

5.7. California adult supporting evidence

I estimate the four-level CITIZEN1 multi-group DiD (Section 3.2). The four-level decomposition immediately reveals a substantive limitation that the pooled public-use specification could not detect: adults without green cards saw large gains in Medi-Cal coverage post-2018 (+9.5 pp, $p < 10^{-6}$) - directionally opposite to a chilling story for the most directly exposed group. A naive read would conclude that chilling did not occur on the coverage margin in California. However, the apparent +9.5 pp gain overlaps almost exactly with California's full-scope Medi-Cal expansions for undocumented adults: ages 19-25 (January 2020), ages 50+ (May 2022), and ages 26-49 (January 2024). The two-window decomposition (Appendix Exhibit A19) confirms the mechanism. In the federal chilling window 2019-2021, the no-green-card Medi-Cal coefficient is +5.4 pp ($p = 0.008$); in the California expansion window 2022-2024 it is +13.4 pp ($p < 0.001$); and the chilling-minus-expansion difference is -8.0 pp ($p < 0.001$), implying that roughly 60-70% of the pooled effect is mechanically the 2022-2024 expansions. In short, the gains were primarily through the California expansions rather than through any easing of the federal chilling pressure.

The cleanest within-state chilling cell available in CHIS is California adults aged 26-49 in the 2019-2021 chilling window - the only adult age band with no full-scope California expansion until January 2024. In this cell, the no-green-card Medi-Cal coefficient is +6.8 pp ($p = 0.006$);

the no-green-card-vs-naturalized contrast is +5.3 pp ($p = 0.055$); and the joint pre-trend Wald test passes ($p = 0.30$). The age-26-49 event study is flat through 2023 and jumps +17.7 pp at 2024, exactly when the full-scope expansion closes for this band. The age-stratified DiD across all four adult age bands (Appendix Exhibit A20) lines up with the California expansion calendar: 19-25 (+10.2 pp, NS), 26-49 (+9.0 pp, $p < 0.001$), 50-64 (+17.2 pp, $p < 0.001$), 65+ (-4.0 pp, NS, Medicare-eligible). The supporting exhibits for this decomposition (the two-window table, the age-stratified table, and the corresponding figures) appear in the appendix (Exhibits A19-A20; Figures A17-A18).

The combination of these results substantiates a clear conclusion: California's broader immigrant-supportive Medi-Cal policy environment was strong enough during 2019-2021 that net Medi-Cal enrollment among adults without green cards rose despite the federal chilling pressure.

5.8. Naturalized-citizen spillover and the fear-based avoidance gradient

Naturalized citizens saw a statistically significant 1.3 percentage point decline in Medi-Cal enrollment ($p = 0.030$; Table 5) even though they are clearly exempt from the public charge rule. As such, a decline in their enrollment is hard to explain as a direct legal response and instead points toward fear operating through household and community networks. The effect does not depend on the no-green-card or green-card estimates and is rather identified in a separate treatment arm relative to U.S.-born citizens.

The restricted-use CHIS data also support a precisely measured fear-based benefit-avoidance gradient. Asked whether they ever decided not to apply for one or more non-cash government benefits because they worried it would disqualify them or a family member from a green card or U.S. citizenship (a question the public-use file does not contain), no-green-card adults are 20.5

percentage points more likely than naturalized citizens to answer yes ($p < 10^{-30}$) and 11.5 percentage points more likely than green-card holders to answer yes ($p < 10^{-14}$). The 12-month-recall variant shows the same gradient: no-green-card vs naturalized = +13.5 pp ($p < 10^{-7}$); green-card vs naturalized = +11.8 pp ($p < 10^{-5}$). These are post-period cross-sectional gradients across citizenship groups, not pre-vs-post DiDs (Section 6.5). These items measure the avoidance decision itself: undocumented adults, and to a smaller extent green-card holders, explicitly report deciding against benefit enrollment because of immigration concerns, at rates 12-20 percentage points above naturalized citizens (Figure 9).

Restricting these items to adults living with at least one child under 18 (the household decision-makers for the citizen children in the national analysis) shows that this fear-based avoidance is concentrated in exactly the study population. Table 6 reports survey-weighted shares of both avoidance items, pooled over 2019-2024, for adults living with children alongside all adults. Among adults living with children, 45.0 percent of those without green cards report ever avoiding a non-cash benefit over immigration concerns, versus 27.6 percent of green-card holders and 13.6 percent of naturalized citizens - a no-green-card-versus-naturalized gradient of 31.3 percentage points, wider than the 28.2-point gradient among all adults (37.9 versus 9.7 percent). In every documentation group, adults living with children report more avoidance than the adult population as a whole. The 12-month-recall item shows the same ordering, and among those who ever avoided, a majority did so within the past year (50.6 percent of adults living with children, rising to 54.2 percent of those without green cards). Adjusting for demographic composition and survey year attenuates the all-adult no-green-card-versus-naturalized gap on the lifetime item to the +20.5-percentage-point regression-adjusted gradient reported above. That the gradient is, if anything, steeper among adults raising children is suggestive that the fear channel

measured here bears directly on the enrollment decisions made for the citizen children the national analysis studies.

County-enforcement heterogeneity. The county-enforcement specification (Section 4.2) provides a within-California enforcement-intensity contrast on the four CHIS adult outcomes. The specification and its limitations are described in Section 4.2.

The headline interaction coefficients are: usual source of care, -8.4 pp ($p < 10^{-4}$); Medi-Cal, -4.5 pp ($p = 0.021$); ED visit, -3.2 pp ($p = 0.049$); forgone care, +1.9 pp ($p = 0.16$, insignificant but right-signed). I point out two notable observations: First, usual source of care is the cleanest single number: large, precisely estimated, and hard to explain by the California-expansion confound that affects the Medi-Cal main effect. Second, naturalized x Post x high-enforcement is null on every outcome (the placebo passes), which strengthens the within-noncitizen-group interpretation of the green-card and no-green-card interactions. The full 24-coefficient table appears in Appendix Exhibit A18.

The within-California enforcement contrast complements the national ACS chilling estimate. Within California, where the pooled Medi-Cal coefficient is expansion-dominated (Section 5.7), the differential by ICE-enforcement intensity is consistent with chilling on access (usual source of care) and on coverage uptake among the most directly exposed group. The result should be read as a robustness/heterogeneity check rather than a co-headline finding; these limitations are documented in Section 6.5 and Appendix Exhibit A18.

5.9. Administrative vulnerability: documentation demands

The CHIS documentation-demand item (AL19V2) asks, “When you applied for health insurance, were you asked for a Social Security number or citizenship documentation?” Because the item is

fielded only to foreign-born respondents beginning in 2019 (Section 4.2), the estimates below are within-immigrant-universe gradients with naturalized citizens as the reference group.

Among 2019-2024 CHIS adult respondents not born in the United States ($N = 33,464$), 18.3 percent of pooled noncitizens and 10.5 percent of naturalized citizens report having been asked for a Social Security number or citizenship documentation when seeking coverage - a 7.8 percentage-point raw gap. The corresponding regression-adjusted gradients are 5.5 to 9.2 percentage points depending on the covariate set (Appendix Exhibit A21). The within-immigrant gradient is materially smaller than what an inclusive-universe specification would imply, but it is still large in policy terms: among adults seeking the same coverage in the same state in the same years, noncitizens are 5 to 9 percentage points more likely than naturalized citizens to be asked for documents that tie the encounter to immigration status.

Two interpretive points follow: First, the within-immigrant-universe gap is a measurement of differential administrative scrutiny among populations who are similarly situated with respect to the actual eligibility test (both are documented immigrants; both are seeking the same coverage). The gap therefore cannot be attributed to objective differences in eligibility complexity; it reflects how the enrollment system handles immigrant applicants differently based on factors the eligibility worker can or cannot observe. Second, the gap is a post-period gradient only (Section 6.5). What the gap does establish is that the administrative environment in which chilling effects operate is structurally more burdensome for noncitizens, and that this differential burden is large enough to plausibly compound the fear channel even when fear itself is not actively rising.

6. Discussion

6.1. Main interpretation

In the national analysis, citizen children in mixed-status families saw a 2.0 percentage point decline in Medicaid enrollment after the September 2018 public charge announcement ($p < 0.001$), and enrollment has not recovered: effects persisted through the policy reversal period and deepened during the 2023-2024 Medicaid unwinding. Declines were primarily among families with a noncitizen parent (-1.8 pp relative to the U.S.-born-parent baseline, $p = 0.003$) rather than across immigrant-connected families broadly, and declines were sharper outside inclusive states (-2.6 pp excluding the seven jurisdictions that extended coverage to undocumented adults, $p < 0.001$). In California, naturalized citizens saw a 1.3 percentage point Medi-Cal decline ($p = 0.030$), and the restricted-use CHIS data identify large fear-based avoidance gradients: adults without green cards are 20.5 percentage points more likely than naturalized citizens to report having declined a non-cash benefit because of immigration concerns, a gradient that is, if anything, steeper among adults living with children. The effect was not just an announcement shock that dissipated once the policy was reversed. It persisted for years and compounded with subsequent administrative disruption during the Medicaid unwinding.

The national ACS results carry the main weight because they directly track the population for whom the theory most clearly predicts a coverage response. The 2.0 percentage point decline is of a significant policy magnitude, and presents a clear picture that children legally entitled to Medicaid lost coverage due to parents facing a difficult choice between providing healthcare to their family and avoiding needed assistance due to fear of elevated immigration risk.

The CHIS results add two things the ACS cannot: evidence that spillovers reached naturalized citizens, and direct evidence on the administrative environment. The naturalized-citizen result is particularly important for mechanism identification.

6.2. Relation to prior literature

The magnitude of the preferred ACS estimate (2.0 pp) is broadly consistent with prior work. Barofsky et al. (2020) estimated effects in roughly the 2-4 percentage point range for the most exposed counties, implying approximately 260,000 fewer children enrolled. My estimate is somewhat smaller as an absolute count, but the designs are not directly comparable. In that sense, I interpret my result less as a contradiction and more as a targeted estimate of the child-level margin.

The welfare reform precedent is also informative. Kandula et al. (2004) estimated a 3 percentage point decline after PRWORA, a magnitude comparable to the estimates here. The parallel suggests that chilling effects of this size are a recurring feature of immigration-health policy interaction, not something unique to the 2019 rule. The enforcement literature provides additional context: Watson's (2014) finding that enforcement accounted for 75% of the noncitizen Medicaid decline shows how much the policy environment, as distinct from formal eligibility rules, shapes enrollment decisions. Alsan and Yang's (2024) Secure Communities estimates showed similarly large spillovers to citizen households, with effects most consistent with network transmission of fear.

The naturalized-citizen result complements the network-mediated chilling literature. The finding that naturalized citizens reduced Medi-Cal enrollment is consistent with the same network mechanism identified by Alsan and Yang: Fear does not stop at legal boundaries; it spreads through families, social networks, and community institutions. The results here fit within this

tradition but study a different type of policy shock: one that explicitly linked benefit use to immigration consequences rather than indirectly raising the perceived cost of government interaction through enforcement activity.

6.3. Mechanisms

While my research design cannot directly test specific mechanisms, I argue that the pattern of my results is most consistent with the following conclusion: the public charge rule generated fear that spread through immigrant communities via media coverage, word of mouth, and community-based organization warnings. That fear raised the perceived cost of enrolling in Medicaid for families who had, or believed they had, members vulnerable to immigration consequences. The fear was overinclusive: it reached populations (citizen children, naturalized citizens) that the rule did not formally target, because the rule's complexity made it hard for families to assess their actual risk. The fear interacted with a pre-existing administrative environment in which immigrants face systematically greater documentation demands when seeking coverage, and the combination of elevated fear and heightened bureaucratic scrutiny created a deterrent sufficient to reduce enrollment among eligible populations.

The persistence after reversal is consistent with the re-engagement-cost and option-value dynamics of Section 2.5, amplified by an information asymmetry: news of the rule's expansion likely reached immigrant communities more effectively than news of its reversal, and the prospect of future policy changes - which subsequently materialized - kept families away.

Direct measurement of fear-based avoidance. A distinctive contribution of this paper is that the restricted-use CHIS analysis provides a direct, individual-level measurement of the fear-based benefit-avoidance channel that the chilling-effects literature has otherwise inferred only indirectly. The AL99/AL104 gradients are reported in Section 5.8.

These coefficients matter for two reasons: First, they directly quantify the household-level decision that produces the population-level coverage patterns – prior literature has documented that immigrants disenroll, that they delay care, and that they shift to self-pay channels - all observable downstream consequences of an unobserved decision to forgo benefit enrollment. The AL99/AL104 gradient measures the decision itself, in a question that explicitly names the immigration channel. The link to the national child analysis is tightest among adults living with children (Section 5.8, Table 6) - the household channel through which a policy aimed at adults reaches children's coverage.

Second, the gradient's structure across the four citizenship groups identifies a pattern that the prior pooled-noncitizen literature could not detect: Green-card holders avoid benefits at rates substantially closer to undocumented adults than to naturalized citizens - the +11.8 pp green-card-vs-naturalized 12-month gap is comparable in magnitude to the +13.5 pp undocumented-vs-naturalized gap. The implication is that the chilling effect operates along a documented-immigrant-status spectrum, not as a binary split between exposed (noncitizen) and unexposed (citizen). Even fully documented green-card holders explicitly forgo benefits because of immigration concerns at rates an order of magnitude above U.S.-born citizens. The chilling effects literature has long hypothesized that fear spreads broadly through immigrant communities; the AL99/AL104 gradient is the first quantification I am aware of that ranks this spread across well-defined documentation tiers.

The implication for policy design is that interventions targeting only the formally affected population (adjustment-of-status applicants) will substantially under-correct for the chilling effect. Effective outreach requires reassurance across the full documented-immigrant-status spectrum, including green-card holders and naturalized citizens with mixed-status family

members. I argue the naturalized-citizen Medi-Cal spillover (-1.3 pp) reinforces this: even adults at the top of the documentation hierarchy responded to the rule's signal.

6.4. Policy implications

Immigration policy as health policy. The most fundamental implication is that immigration policy can function as de facto health policy: a rule directed at immigration adjudication produced measurable coverage losses among U.S. citizen children who were never legally subject to it. The cost-benefit analysis for immigration rules of this type should include health coverage effects as foreseeable consequences, not unanticipated side effects.

Administrative environment and enrollment infrastructure. The documentation-demand gradient shows that enrollment systems treat immigrants substantively differently from native-born citizens. This administrative gap creates the conditions under which fear can translate into coverage loss. If enrollment systems were designed to minimize documentation demands for eligible populations (for example, by not asking parents about their own immigration status when processing children's Medicaid applications), the chilling effect of policies like the public charge rule might be attenuated. California's relatively protective policy environment likely muted the direct adult coverage response observed in CHIS; that makes the administrative-vulnerability gradient more, not less, important for understanding why even protective states remain susceptible to chilling.

Persistence of fear. Reversing a restrictive and/or punitive rule does not quickly erase its effects, especially when subsequent administrative shocks interact with already-elevated fear. The persistence through reversal and deepening during the unwinding suggest that policymakers should anticipate lasting behavioral consequences from restrictive immigration policies. Outreach, community education, and proactive re-enrollment efforts may be necessary to

counteract residual effects of fear-generating policies, even after those policies are formally rescinded – these repeated efforts perhaps allow for the “slow melting” that is required to reverse the chilling.

Current policy relevance. Under the second Trump administration’s refinalized public charge rule (Section 2.3), these results suggest a revival would produce enrollment losses at least as large as the 2019 episode, and possibly larger if prior fear has not fully dissipated. The asymmetry between scaring families away from programs and bringing them back implies that the cumulative effects of repeated restrictive episodes may exceed any single episode. State-level protective policies may partially buffer federal chilling effects (Section 5.4).

6.5. Limitations

Potential limitations include treatment group heterogeneity, pre-trend sensitivity, and post-period contamination; I discuss each below in detail.

Treatment group heterogeneity. The ACS mixed-status definition pools families with very different vulnerability levels, from those with undocumented parents (highest risk) to lawful permanent residents (lower risk) to temporary visa holders. This pooling likely attenuates estimates relative to effects among the most vulnerable families. CHIS partially resolves this pooling by separating green-card holders from noncitizens without green cards, but California’s contemporaneous full-scope Medi-Cal expansions preclude a clean within-state coverage-chilling estimate (Section 5.7). What the California analysis identifies cleanly is the naturalized-citizen Medi-Cal spillover and the fear-based avoidance gradients (Section 5.8). The California children mixed-family CHIS specification (Appendix Exhibit A22) is null on coverage, which is consistent with California’s pre-window expansion of full-scope Medi-Cal to undocumented

children (May 2016) and the small CHIS California children sample (N = 9,607 below 200% FPL vs the ACS full child sample of 6.5 million), not with absence of national chilling.

Extended pre-period pre-trend issues and the ACA overlap. Over the 2014-2024 window, the preferred below-200% FPL Medicaid event study does not reject parallel pre-trends (joint $p = 0.14$), but the extended leads are uniformly positive and several secondary specifications fare worse (Section 5.2). The four pre-treatment years strengthen the design relative to a two-lead window, but 2014 and 2015 sit inside the ACA's welcome-mat window (Sections 2.5, 5.2), and the joint p -value is materially lower than in the two-lead window. The summary judgment is that Medicaid enrollment in the below-200% FPL sample is the robust outcome-sample combination.

State-environment heterogeneity is suggestive, not decisive. The formal national triple-DiD interaction is underpowered and not statistically significant, and the DROP-7 specification's joint pre-trend test rejects over the extended window, so I report state-environment heterogeneity (Section 5.4) as a robustness/heterogeneity finding rather than a co-primary specification. Future research with finer state-policy controls (sanctuary signaling, navigator infrastructure, state-funded coverage uptake) would help distinguish among candidate channels.

County-enforcement heterogeneity is AOR-level, not within-AOR county-level. As described in Section 4.2, the enforcement contrast is a Southern-versus-Northern California AOR-pair partition, with 2015-2021 values back-filled from 2022, and pre-trend tests on the interaction term are not separately estimated. The county-enforcement results in Section 5.8 and Appendix Exhibit A18 are therefore robustness evidence, not a co-primary specification.

AL99 and AL104 are post-period cross-sections, not DiDs. The two restricted-use fear-avoidance items (AL99 and AL104) are absent from the 2015-2018 adult questionnaires and present in 2019-2024, which I confirmed against the published UCLA CHIS questionnaire

archive. The reported coefficients are post-period cross-sectional differences across CITIZEN1 groups, not pre-vs-post DiDs, and as such cannot have causal interpretations. They cannot test whether the gradient changed with the public charge rule, because no pre-period observations exist. The same caveat applies to the documentation-demand variable, which was first collected in 2019.

Different estimands across ACS and CHIS. The ACS and CHIS analyses are intentionally complementary, not identical. The ACS targets Medicaid-eligible citizen children, while the CHIS analysis studies adults in one state and is used mainly to probe spillovers and administrative vulnerability. Readers should not interpret the California estimates as a second version of the national child analysis. Pairwise event-study diagnostics reveal no missingness, empty-cell, or saturation problem in the California data, but the naturalized-versus-U.S.-born Medi-Cal path is less clean than the static DiD estimate alone, so I treat the naturalized-citizen result as suggestive supporting evidence on spillovers rather than as a second headline causal estimate.

Post-period contamination. The post-announcement period includes multiple overlapping shocks: the public charge rule, COVID-19, the continuous enrollment provision, and the Medicaid unwinding. DiD nets out common shocks affecting both groups equally, but any shock differentially affecting mixed-status families could bias estimates. The pandemic disproportionately affected immigrant communities through higher infection rates, economic disruption, and differential access to services. I address this by 1) splitting the post period into pre-unwinding and unwinding windows, and 2) re-estimating with alternative treatment timing; these provide partial leverage but cannot fully resolve the confounding.

CHIS utilization outcomes. The positive relative changes in ED use and usual source of care are difficult to reconcile with a chilling-effect story and likely reflect COVID-era trends and California's Medi-Cal expansions. I present them for completeness but do not build the argument on them.

External validity. The CHIS analysis is specific to California, which has a large immigrant population and relatively protective policies. Chilling effects may be larger in less protective states or smaller where noncitizen populations are less prevalent.

6.6. Future research

I propose several extensions that would strengthen the evidentiary base of this paper. Linking the ACS or CHIS analysis with state-level administrative Medicaid enrollment data would allow validation of the survey-based coverage measures and a check on whether the estimated enrollment declines appear in administrative records. Exploiting state-level variation in protective policies (administrative firewalls, state-funded coverage, outreach programs) would provide evidence on whether and how states can buffer federal chilling effects. Studying the current policy cycle (2025 onward) in real time would test whether the chilling effects of the earlier episode compound or dissipate, and whether prior exposure makes immigrant communities more or less sensitive to new threats. Lastly, following the children who lost Medicaid coverage during this period over time would provide evidence on the long-run health and educational consequences of coverage disruption, extending the literature on the life-course effects of childhood Medicaid (Miller and Wherry, 2019).

7. Conclusion

The public charge episode shows how a policy targeted at immigration status can reshape health coverage for populations well outside its formal legal scope. The mechanism is not legal exclusion - the populations most affected by the chilling effect were never subject to the rule – but rather, fear: fear amplified by policy complexity, transmitted through immigrant networks, and made possible by an administrative environment that scrutinizes immigrant applicants more intensively than native-born citizens.

I estimate that low-income U.S. citizen children in mixed-status families experienced a 2.0 percentage point decline in Medicaid enrollment after the public charge rule's announcement, an effect that persisted through the rule's reversal and deepened during the Medicaid unwinding. California provides supporting evidence that chilling fears spilled over to naturalized citizens, a group with no direct legal vulnerability to the rule, and that immigrant groups face a markedly more burdensome documentation environment when seeking health coverage. The California adult results are not as decisive as the national ACS child analysis, but they clarify the spillover and administrative channels through which the rule likely operated.

These findings carry implications for both immigration and health policy. For immigration policy, they show that the collateral effects of restrictive rules on health coverage are substantial, measurable, and persistent. For health policy, they show that enrollment infrastructure and administrative design can either amplify or attenuate the chilling effects of policy shocks. For both, they point to a basic asymmetry: it is easier to generate fear than to undo it.

The public charge rule has been formally reversed, but its effects have not been fully erased. Any future reinstatement of similar restrictions should be evaluated more holistically as child health policy and Medicaid policy, not just as immigration policy.

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Tables

Table 1. ACS Children Descriptive statistics

Variable	Mixed-Status	Citizen-Only	Std. Diff.
N (pre-period)	373,687	2,677,169	-
Age (mean)	8.24	9.04	-0.15
Female	0.489	0.489	0.00
Below 200% FPL	0.584	0.335	0.51
Medicaid (pre-period)	0.602	0.337	0.55
Uninsured (pre-period)	0.071	0.037	0.15
Any coverage (pre-period)	0.929	0.963	-0.15

Notes: Pre-period (2014-2018) weighted means for the full citizen-child sample. Standardized differences are reported to characterize level differences between treatment and control groups. The full 2014-2024 sample contains 754,547 mixed-status and 5,727,480 citizen-only children; the preferred below-200% FPL analysis sample contains 389,899 mixed-status and 1,687,970 citizen-only children.

Table 2. CHIS Data-source map

CHIS specification	File	Required variables
Three-group DiD on Medi-Cal, ED, usual source, forgone care	Public-use	CITIZEN2, INSMC, ER, USUAL, FORGO
Naturalized x Post on Medi-Cal (-1.3 pp), Section 5.8	Public-use	CITIZEN2, INSMC
Documentation-demand, immigrant-only universe, Section 5.9	Public-use	AL19V2 (immigrant universe), CITIZEN2
Lifetime fear-avoidance (AL99-equivalent), Section 5.8	Public-use	CITIZEN2, AL99 indicator
Four-level CITIZEN1 multi-group DiD (no GC vs GC vs nat vs US-born)	Restricted	CITIZEN1
Two-window decomposition (federal chilling vs CA expansion windows)	Restricted	CITIZEN1, INSMC
Age-26-49 within-CA “cleanest cell” event study	Restricted	CITIZEN1, INSMC
Fear-avoidance shares, adults living with children vs. all adults (Section 5.8)	Restricted	CITIZEN1, KIDCNT, AL99, AL104
12-month fear-avoidance gradient (AL104), Section 5.8	Restricted	CITIZEN1, AL104
County-enforcement heterogeneity, Appendix Exhibit A18	Restricted	CITIZEN1, FIPS_CNT

Notes: Maps each California specification to the CHIS data product that generates it. The pooled 2015-2024 adult sample (N = 220,925) is identical across the public-use and restricted-use files; the restricted-use file adds confidential variables and the finer four-level citizenship taxonomy. Discussed in Section 3.2.

Table 3. ACS Children Change in coverage over time, DiD estimates

Sample	Outcome	Beta	SE	p	N
Below 200% FPL	Medicaid	-0.020	0.005	<0.001	2,077,869
Below 200% FPL	Any public coverage	-0.020	0.004	<0.001	2,077,869
Below 200% FPL	Uninsured	+0.008	0.005	0.115	2,077,869

Sample	Outcome	Beta	SE	p	N
Full sample	Medicaid	-0.005	0.005	0.345	6,482,027

Notes: Estimation window 2014-2024. The below-200% FPL Medicaid specification is the

paper's preferred ACS design. The full-sample Medicaid estimate is secondary; full-sample uninsured and any-coverage results are not reported as evidence because the full-sample uninsured event study fails the joint pre-trend test over the extended window ($p < 0.001$; Appendix Exhibit A4).

Table 4. ACS Children Change in Medicaid over time, Expansion vs Non-Expansion States

Specification	Term	Beta	SE	p	N
DROP-7 (excl. CA, IL, OR, NY, CO, WA, DC)	Mixed x Post	-0.026	0.007	<0.001	1,541,715
National triple-DiD	Mixed x Post (non-expansion baseline)	-0.026	0.007	<0.001	2,077,869
National triple-DiD	Mixed x Post x Expansion-state	+0.008	0.008	0.341	2,077,869

Notes: Below-200% FPL sample, Medicaid outcome, 2014-2024. "Expansion state" indicates the seven jurisdictions that extended Medicaid or near-equivalent state-funded coverage to undocumented adults during the paper window (CA, IL, OR, NY, CO, WA, DC). The DROP-7 event study's joint pre-trend test rejects over the extended window ($p = 0.019$), driven by a positive 2014 lead (+2.1 pp, $p = 0.015$); the state-environment results are reported as heterogeneity evidence, not as a co-primary specification. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2024).

Table 5. CHIS Adults Change in outcomes over time, multi-group DiD

Outcome	Noncitizen x Post	p	Naturalized x Post	p	N
Medi-Cal	-0.000	0.979	-0.013	0.030	220,925
ED visit	+0.055	<0.001	+0.050	<0.001	220,925
Usual source	+0.054	<0.001	+0.007	0.368	220,925

Outcome	Noncitizen x Post	p	Naturalized x Post	p	N
Forgone care	-0.004	0.616	-0.012	0.027	220,925

Notes: Reference group is U.S.-born citizens. Robust standard errors are the primary CHIS

inference. The naturalized-citizen Medi-Cal estimate is the paper's preferred California spillover result. CHIS descriptive statistics appear in Appendix Exhibit A1; the documentation-demand gradients appear in Appendix Exhibit A21.

Table 6. CHIS Adults Fear-based benefit avoidance, Adults Living With Children vs All Adults, 2019-2024

Panel A. Ever avoided a non-cash benefit over immigration concerns (AL99)

Documentation group	Adults living with children: share (SE) [95% CI], n	All adults: share (SE) [95% CI], n
All foreign-born groups	25.5% (0.7) [24.1, 27.0], 9,965	17.7% (0.3) [17.0, 18.3], 33,464
Naturalized citizen	13.6% (0.6) [12.5, 14.8], 6,032	9.7% (0.3) [9.1, 10.2], 24,085
Green-card holder	27.6% (1.5) [24.7, 30.5], 2,122	22.1% (0.8) [20.6, 23.6], 5,499
No green card	45.0% (1.6) [41.7, 48.2], 1,811	37.9% (1.0) [35.9, 40.0], 3,880

Panel B. Among those who ever avoided, avoided in the past 12 months (AL104)

Documentation group	Adults living with children: share (SE) [95% CI], n	All adults: share (SE) [95% CI], n
All foreign-born groups	50.6% (1.3) [48.0, 53.2], 1,988	45.5% (1.0) [43.5, 47.5], 4,198
Naturalized citizen	41.0% (2.4) [36.3, 45.7], 666	34.0% (1.7) [30.7, 37.3], 1,737
Green-card holder	53.9% (2.9) [48.2, 59.5], 500	48.9% (2.1) [44.8, 53.0], 1,076
No green card	54.2% (2.2) [49.9, 58.4], 822	52.5% (1.6) [49.3, 55.7], 1,385

Notes: Survey-weighted shares with jackknife replicate-weight standard errors (in parentheses),

95% confidence intervals (in brackets), and unweighted denominators (n), pooled over CHIS

2019-2024, the survey years in which the AL99 and AL104 items are fielded. "Adults living with children" are respondents in households with at least one child under 18 (CHIS KIDCNT >= 1).

The AL99 and AL104 items are asked only of foreign-born respondents, so U.S.-born citizens are not shown and the "all foreign-born groups" row is the applicable full-sample total. Panel B is conditional on AL99 = yes (ever avoided), i.e., the share of those who ever avoided who did so within the past year. Author's calculations from restricted-use CHIS adult output returned by the UCLA DAC.

Figures

Figure 1. Legal scope and spillover.

The 2019 public charge rule: who it legally touched vs. who changed behavior

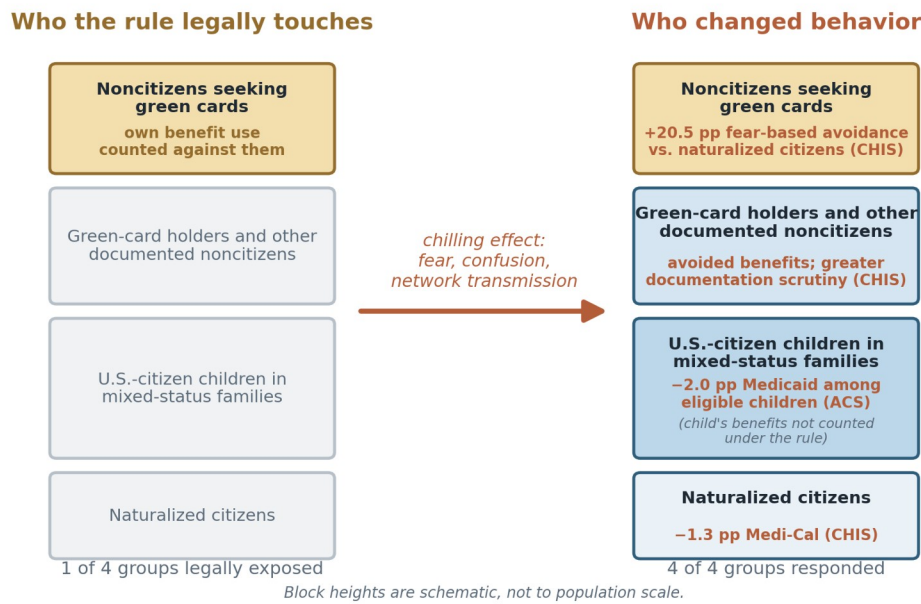


Figure 1

Legal scope of the public charge rule and the exempt populations its chilling effects reached.

Figure 2. Evidence structure.

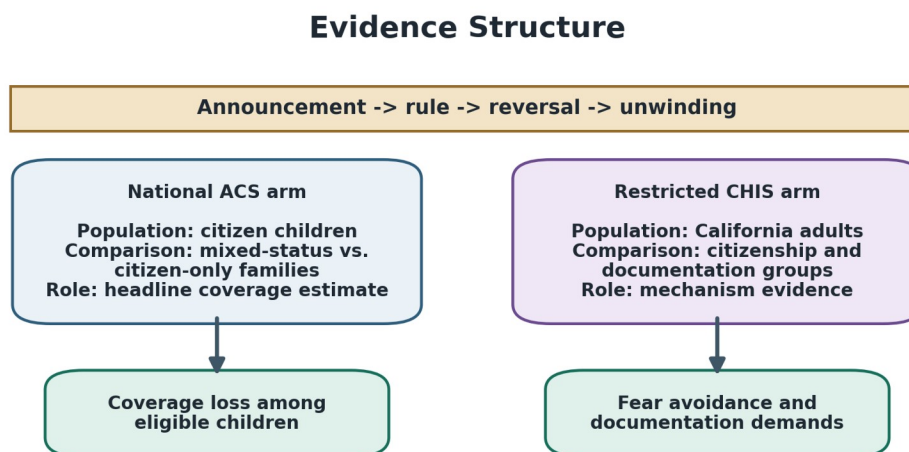


Figure 2

How the national ACS child analysis and the supporting California CHIS analysis fit together.

Figure 3. Public charge policy timeline.

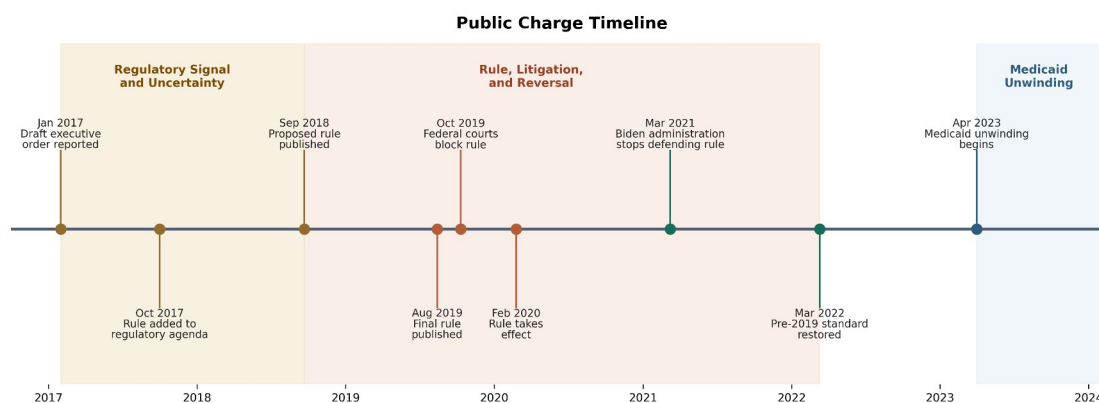


Figure 3

Policy timeline for the public charge rule and the Medicaid unwinding, from the 2017 regulatory buildup through the formal restoration of the pre-2019 standard and the later unwinding shock.

Figure 4. Exposure framework.

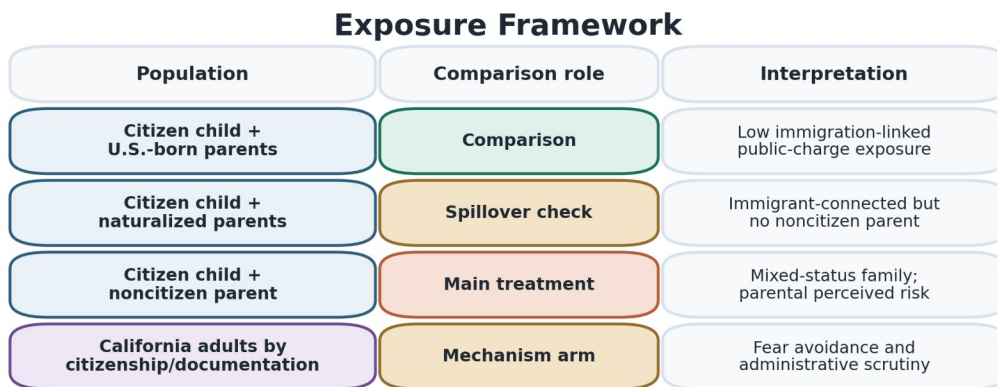


Figure 4

Household-level exposure framework for the ACS design: which family types face direct legal exposure versus perceived risk.

Figure 5. Raw Medicaid enrollment trends, below 200% FPL.

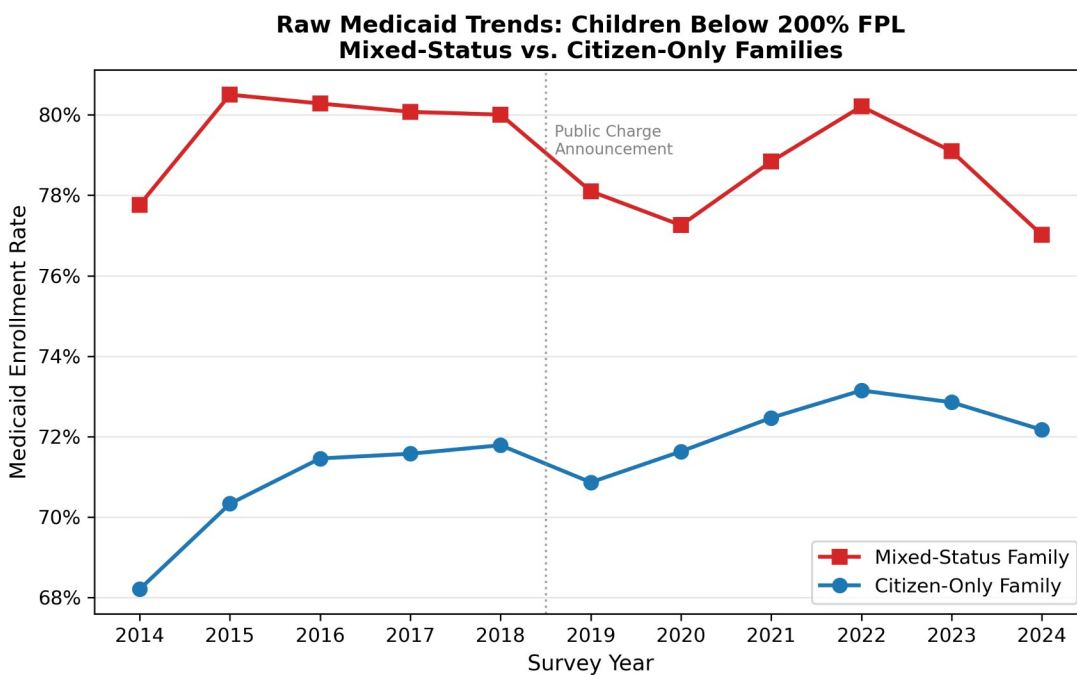


Figure 5

Weighted mean Medicaid enrollment for U.S. citizen children in mixed-status versus citizen-only families, below 200% FPL, 2014-2024; a gap opens after the September 2018 announcement and persists through 2024.

Figure 6. Event study, Medicaid enrollment, below 200% FPL.

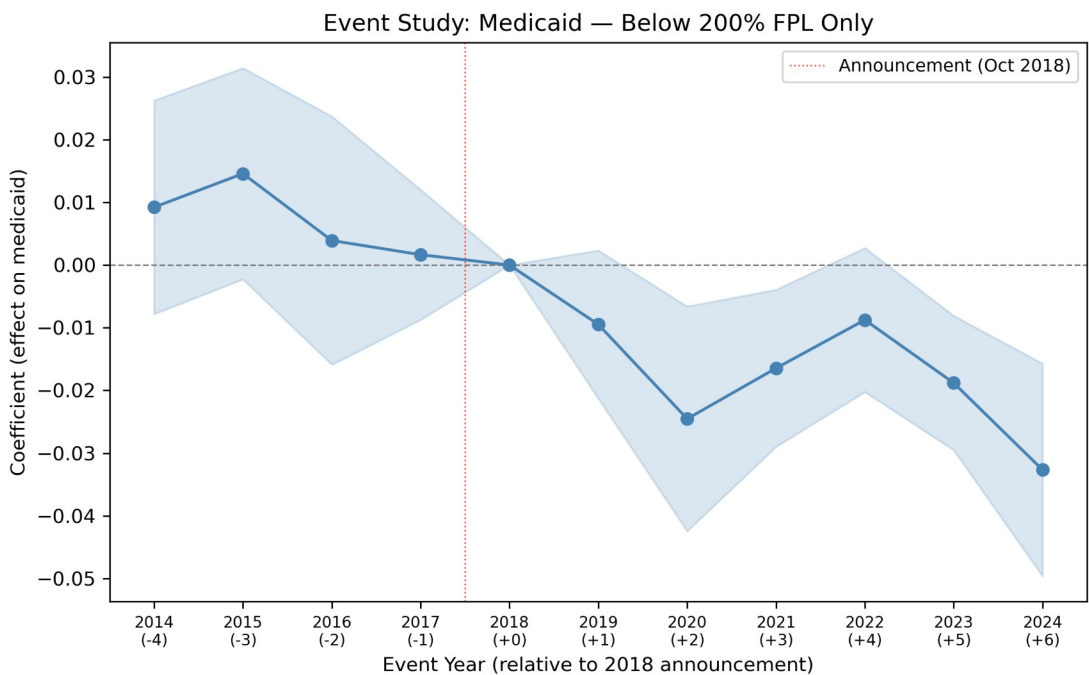


Figure 6

Four pre-announcement leads (2014-2017) are individually insignificant and jointly indistinguishable from zero (joint Wald $p = 0.14$), though uniformly positive; the post-period path turns negative after the announcement and remains negative through 2024.

Figure 7. Naturalized-parent decomposition event study.

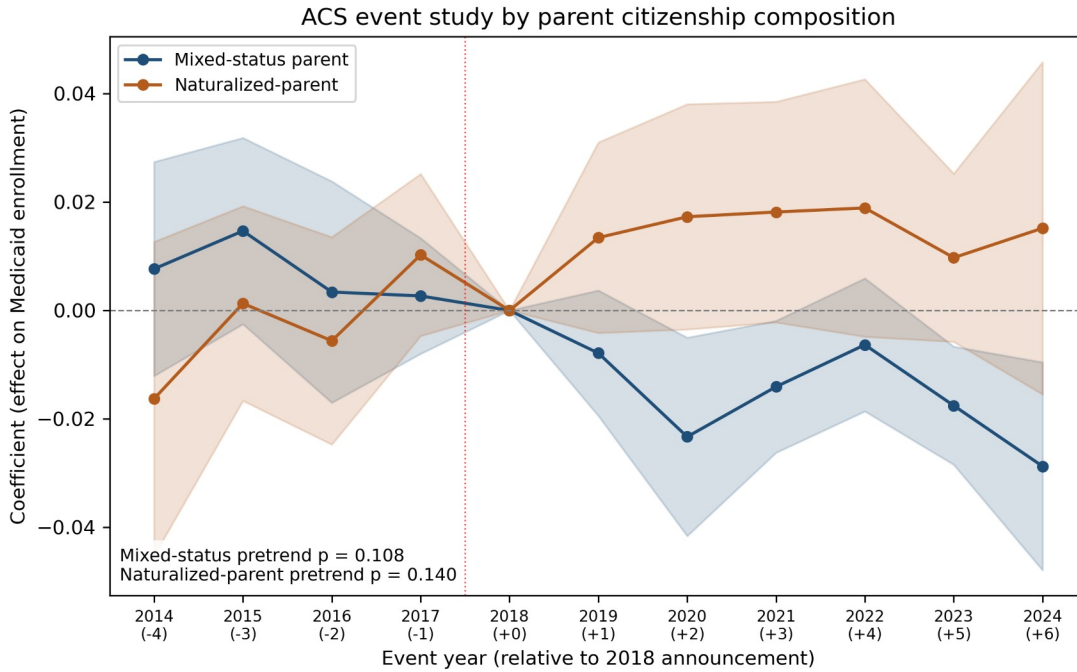


Figure 7

Event studies for mixed-status and naturalized-parent families relative to the U.S.-born-parent baseline; the chilling effect concentrates in households with a noncitizen parent.

Figure 8. DROP-7 event study.

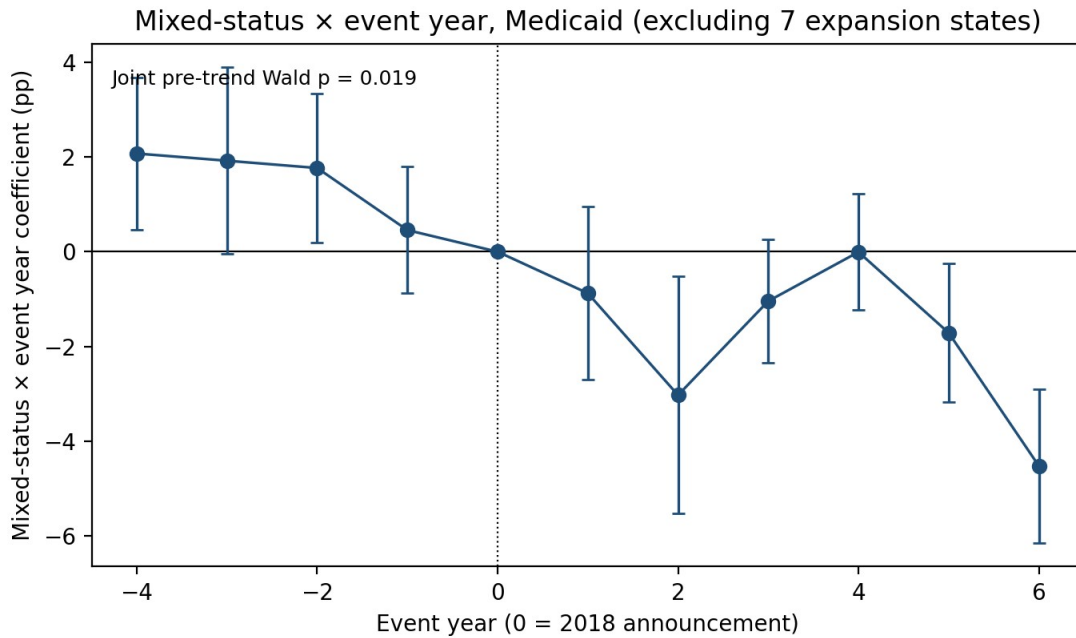


Figure 8

Event study excluding the seven jurisdictions that extended coverage to undocumented adults during the paper window; the chilling estimate is larger outside inclusive states.

Figure 9. Fear-based benefit-avoidance gradient (CHIS restricted-use items).

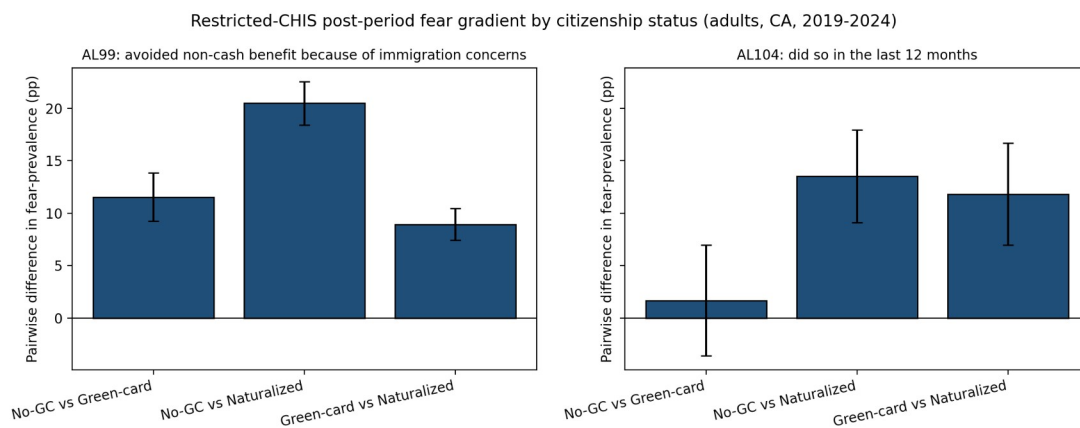


Figure 9

Survey-weighted fear-based avoidance shares by documentation group for the lifetime (AL99) and 12-month (AL104) items, CHIS 2019-2024.

Supplementary Appendix

Appendix Exhibit A1: Descriptive Statistics - CHIS Sample

Variable	US-Born	Naturalized	Noncitizens
N	168,525	35,656	16,744
Medi-Cal	0.224	0.254	0.057
ED visit	0.205	0.169	0.169
Usual source	0.857	0.862	0.715
Forgone care	0.103	0.064	0.079
Asked SSN/cit	0.000	0.073	0.117
Age (mean)	46.5	53.3	42.2
Female	0.511	0.524	0.497
Below 200% FPL	0.257	0.363	0.570

Appendix Exhibit A2: ACS Full-Sample Results (All Income Levels)

Specification	Outcome	Beta	SE	p	N
Basic DiD	Medicaid	-0.005	0.005	0.35	6,482,027
Triple-DiD	Medicaid	-0.004	0.007	0.62	6,482,027
(mixed x high-NC x post)					
Drop COVID (20-21)	Medicaid	-0.003	0.006	0.63	5,429,557
COVID interaction	Medicaid	-0.003	0.006	0.61	6,482,027

Notes: Full-sample results presented as secondary specifications, estimated on the 2014-2024

window. The attenuated Medicaid effect (-0.5 pp, $p = 0.35$) is consistent with dilution from

higher-income families for whom Medicaid is not the relevant coverage margin. Full-sample

uninsured and any-coverage DiD estimates are not reported as evidence because the full-sample

uninsured event study rejects the joint pre-trend test decisively over the extended window ($p <$

0.001; see Exhibit A4).

Appendix Exhibit A3: Event-Study Coefficients - Medicaid, Full Sample

Year	Beta	SE	p
2014 (-4)	-0.006	0.009	0.48
2015 (-3)	+0.005	0.008	0.57
2016 (-2)	+0.007	0.008	0.34

Year	Beta	SE	p
2017 (-1)	+0.005	0.005	0.33
2018 (ref)	0	-	-
2019 (+1)	-0.002	0.005	0.66
2020 (+2)	-0.011	0.004	0.016
2021 (+3)	+0.003	0.007	0.63
2022 (+4)	+0.008	0.004	0.027
2023 (+5)	+0.004	0.003	0.22
2024 (+6)	-0.006	0.007	0.38

Appendix Exhibit A4: Joint Wald Tests of Pre-Period Event-Study Coefficients

Analysis	Outcome	N pre-periods	Chi-square	p
ACS (below 200% FPL)	Medicaid	4	6.92	0.140
ACS (full sample)	Uninsured	4	35.01	<0.001
ACS (full sample)	Any coverage	4	35.01	<0.001
CHIS	Medi-Cal	3	1.11	0.77
CHIS	ED visit	3	1.20	0.75
CHIS	Usual source	3	4.13	0.25
CHIS	Forgone care	3	3.34	0.34

Notes: Joint Wald tests use the estimated covariance matrix of the pre-period event-study

coefficients rather than an approximation that ignores covariance across leads. ACS tests use the four extended pre-treatment leads (2014-2017). The primary specification (below 200% FPL, Medicaid) does not reject parallel pre-trends ($p = 0.140$), though the joint p -value is materially lower than in the two-lead (2016-2017) window ($p = 0.92$) because the 2014-2015 leads are positive; Section 5.2 discusses this pattern and its interpretation. The full-sample uninsured (and its mirror-image any-coverage) pre-trend test rejects decisively ($p < 0.001$), so the full-sample uninsurance margin is not interpreted as evidence. All CHIS outcomes pass the joint Wald test.

Appendix Exhibit A5: Placebo Tests - ACS

Test	Outcome	Beta	SE	p	N
Citizen-only x High NC x Post	Medicaid	-0.004	0.005	0.41	5,727,480
Citizen-only x High NC x Post	Uninsured	+0.002	0.002	0.37	5,727,480
Citizen-only x High NC x Post	Any coverage	-0.002	0.002	0.37	5,727,480

Notes: Placebo test using citizen-only families comparing high vs. low noncitizen-share states.

All null, confirming the treatment effect is specific to mixed-status families.

Appendix Exhibit A6: COVID-19 Sensitivity - CHIS

Outcome	Specification	Beta	SE	p	N
Medi-Cal	Drop COVID	-0.003	0.006	0.61	146,774
ED visit	Drop COVID	+0.064	0.011	<0.001	146,774
Medi-Cal	COVID interaction	-0.003	0.006	0.58	185,269
ED visit	COVID interaction	+0.064	0.011	<0.001	185,269

Notes: Secondary two-group CHIS comparisons restricted to noncitizens versus U.S.-born citizens. The preferred California specification in the main text is the multi-group model reported in main-text Table 5.

Appendix Exhibit A7: BRR Variance Estimation - CHIS

Outcome	Coefficient	Beta	BRR		BRR p
			SE	Robust SE	
Medi-Cal	Noncitizen x Post	-0.000	0.001	0.006	0.85
Medi-Cal	Naturalized x Post	-0.013	0.001	0.006	<0.001
ED visit	Noncitizen x Post	+0.055	0.001	0.010	<0.001
ED visit	Naturalized x Post	+0.050	0.001	0.008	<0.001
Usual source	Noncitizen x Post	+0.054	0.002	0.012	<0.001
Usual source	Naturalized x Post	+0.007	0.001	0.008	<0.001
Forgone care	Noncitizen x Post	-0.004	0.001	0.007	<0.001
Forgone care	Naturalized x Post	-0.012	0.001	0.005	<0.001

Notes: BRR SE from 80 CHIS replicate weights for the preferred multi-group CHIS model in main-text Table 5. The BRR SEs are substantially smaller than heteroskedasticity-robust SEs because BRR captures survey sampling variance only, while the robust SEs also absorb misspecification and heteroskedasticity concerns. I therefore retain the robust SEs as the primary CHIS inference.

Appendix Exhibit A8: Alternative Treatment Timing (Post = 2020, Implementation)

Analysis	Outcome	Beta	SE	p	N
ACS	Medicaid	-0.003	0.003	0.41	6,482,027

Analysis	Outcome	Beta	SE	p	N
ACS	Uninsured	+0.005	0.002	0.036	6,482,027
CHIS	Medi-Cal	-0.004	0.006	0.48	185,269
CHIS	ED visit	+0.050	0.010	<0.001	185,269

Notes: CHIS rows report the secondary two-group noncitizen-versus-U.S.-born comparison. The preferred California specification in the main text is the multi-group model reported in main-text Table 5.

Appendix Exhibit A9: Heterogeneity by Poverty Level - CHIS

Subgroup	Outcome	Beta	SE	p	N
Below 200% FPL	Medi-Cal	-0.001	0.009	0.87	-
Above 200% FPL	Medi-Cal	-0.013	0.008	0.11	-
Below 200% FPL	ED visit	+0.065	0.014	<0.001	-
Above 200% FPL	ED visit	+0.048	0.014	<0.001	-

Notes: Secondary two-group CHIS comparisons restricted to noncitizens versus U.S.-born citizens. The preferred California specification in the main text is the multi-group model reported in main-text Table 5.

Appendix Exhibit A10: Unwinding Decomposition

Sample	Period	Outcome	Beta	SE	p	N
Below 200% FPL	Full post (2019-2024)	Medicaid	-0.020	0.005	<0.001	2,077,869
Below 200% FPL	Pre-unwinding (2019-2022)	Medicaid	-0.017	0.005	0.001	2,077,869
Below 200% FPL	Unwinding (2023-2024)	Medicaid	-0.028	0.008	<0.001	2,077,869
Full sample	Pre-unwinding (2019-2022)	Medicaid	-0.004	0.004	0.287	6,482,027
Full sample	Unwinding (2023-2024)	Medicaid	-0.005	0.007	0.468	6,482,027

Sample	Period	Outcome	Beta	SE	p	N
Below 200% FPL	Pre-unwinding (2019-2022)	Uninsured	+0.006	0.005	0.180	2,077,869
Below 200% FPL	Unwinding (2023-2024)	Uninsured	+0.010	0.005	0.070	2,077,869

Notes: Split-period unwinding decomposition, 2014-2024 window. The below-200% FPL

Medicaid rows are the split-period estimates discussed in Section 5.5. Consistent with the primary specification, effects are attenuated in the full sample. The below-200% FPL uninsured result shows a marginally significant unwinding-period effect ($p = 0.070$), directionally consistent with the Medicaid findings. Author's calculations from American Community Survey microdata (IPUMS USA, 2014-2024).

Appendix Exhibit A11: Earlier Treatment Timing Sensitivity

Timing	Sample	Outcome	Beta	SE	p
Baseline: post=2019 + (NPRM)	Below 200% FPL	Medicaid	-0.020	0.005	<0.001
Alt A: post=2018 + (Reg Agenda)	Below 200% FPL	Medicaid	-0.018	0.007	0.012
Alt A: post=2018 + (Reg Agenda)	Below 200% FPL	Uninsured	+0.009	0.005	0.086
Alt A: post=2018 + (Reg Agenda)	Full sample	Medicaid	-0.006	0.006	0.302
Alt B: post=2017 + (Leaked EO)	Below 200% FPL	Medicaid	-0.017	0.010	0.086
Alt B: post=2017 + (Leaked EO)	Below 200% FPL	Uninsured	+0.008	0.005	0.150
Alt B: post=2017 + (Leaked EO)	Full sample	Medicaid	-0.007	0.008	0.366

Notes: Sensitivity to alternative treatment timing reflecting the public charge rule’s emergence before the October 2018 NPRM, estimated on the 2014-2024 window. A leaked draft executive order on public benefits was reported in January 2017 (Washington Post); DHS placed the rule on the Unified Regulatory Agenda in October 2017. Alt A (post=2018+) remains significant for below-200% FPL Medicaid, suggesting anticipation effects. Alt B (post=2017+) is directionally consistent but attenuated and marginally significant, consistent with gradual onset. The event study centered on 2016 (Exhibit A12; Appendix Figure A11) shows a flat 2017 coefficient, with effects emerging in 2018 and deepening from 2020.

Appendix Exhibit A12: Event-Study Coefficients - Medicaid, Below 200% FPL (Reference: 2016)

Year	Event Year	Beta	SE	p
2014	-2	+0.005	0.006	0.34
2015	-1	+0.011	0.006	0.063
2016 (ref)	0	0	-	-
2017	+1	-0.002	0.008	0.78
2018	+2	-0.004	0.010	0.70
2019	+3	-0.013	0.012	0.26
2020	+4	-0.028	0.016	0.076
2021	+5	-0.020	0.007	0.007
2022	+6	-0.013	0.007	0.070
2023	+7	-0.023	0.009	0.012
2024	+8	-0.037	0.017	0.033

Notes: Event study with 2016 as the reference year to examine whether effects began before the NPRM; the extended extract adds 2014 and 2015 as pre-2016 leads. The 2014-2015 leads are small and positive (the 2015 lead is marginally significant, $p = 0.063$), consistent with the gently elevated early leads discussed in Section 5.2. The 2017 coefficient is near zero (-0.2 pp, $p = 0.78$), suggesting no measurable chilling from the leaked EO at the population level. Effects emerge gradually from 2018 and deepen sharply from 2020. This trajectory is consistent with chilling intensifying as the policy moved from leaked draft to formal rulemaking to implementation.

Appendix Exhibit A13: Pairwise CHIS Event-Study Diagnostics

Panel A. Sample support and saturation

Comparison	Treated N	Control N	Missingness in modeled variables	Empty year-by-group cells	Fully saturated event study
Noncitizens vs. U.S.-born citizens	16,744	168,525	No	No	Yes
Naturalized vs. U.S.-born citizens	35,656	168,525	No	No	Yes

Panel B. Medi-Cal pairwise results

Comparison	Pairwise DiD Beta	SE	p	Joint pre-trend p	Average post coefficient
Noncitizens vs. U.S.-born citizens	-0.001	0.006	0.789	0.946	-0.000
Naturalized vs. U.S.-born citizens	-0.013	0.006	0.030	0.171	-0.030

Notes: Pairwise CHIS event-study diagnostic. Panel A summarizes the missingness and cell-support checks; Panel B reports the pairwise DiD and event-study summary for Medi-Cal. Both pairwise comparisons estimate all expected year interactions. The naturalized-versus-U.S.-born event study does not reject the joint pre-trend test, but one lead coefficient is modestly negative, so I treat that California result as supporting rather than definitive evidence. Author's calculations from CHIS adult microdata.

Appendix Exhibit A14: Pairwise CHIS Medi-Cal Event Studies

See Appendix Figure A15. *Notes: Pairwise event-study estimates comparing noncitizens with U.S.-born citizens and naturalized citizens with U.S.-born citizens. Reference year is 2018. The noncitizen series remains close to zero throughout. The naturalized series is negative in most*

post-period years, but one pre-period coefficient is also modestly negative, reinforcing a cautious interpretation of the California spillover result.

Appendix Exhibit A15: Triple-Difference Calibration Check - High Noncitizen-Share States

Coefficient	Beta	SE	p
Mixed-status x High NC share	+0.057	0.014	<0.001
High NC share x Post	-0.004	0.005	0.437
Mixed-status x High NC share x Post	-0.003	0.007	0.703

Notes: Triple-difference calibration check using the full ACS child sample. The null triple interaction shows that the preferred result is not strengthened by additionally interacting exposure with high state-year noncitizen share. I therefore treat this as a weak secondary specification rather than a core identification result.

Appendix Exhibit A16: ACS State-by-Year Fixed-Effects Event-Study Summary

Analysis	Average post coefficient	SE	p	Joint pre-trend p	N
Medicaid, below 200% FPL	-0.019	0.004	<0.001	0.078	2,077,869

Notes: Separate ACS event-study sensitivity with state-by-year fixed effects, estimated on the preferred below-200% FPL sample over 2014-2024. This specification absorbs any state-specific shocks common to mixed-status and citizen-only children within the same year. The average post coefficient is the mean of the 2019-2024 event-study coefficients. Over the extended window the joint pre-trend test is borderline (chi-square = 8.42, $p = 0.078$, four leads), driven by a positive 2014 lead (+1.5 pp, $p = 0.026$); the 2015-2017 leads are individually insignificant.

Appendix Exhibit A17: ACS Event Study with State-by-Year Fixed Effects

See Appendix Figure A16. Notes: State-by-year fixed-effects event-study estimates for Medicaid enrollment among children below 200% FPL, 2014-2024. Reference year is 2018. The 2015-2017 leads are near zero; the 2014 lead is positive (+1.5 pp, $p = 0.026$), and the joint pre-trend test is borderline ($p = 0.078$; Exhibit A16). The post-period path turns negative in 2020 and

remains negative through 2024. This supports the interpretation that the main ACS result is not driven by broad state-level shocks that affect both family types equally within a year, subject to the extended pre-period caveats discussed in Section 5.2.

Appendix Exhibit A18: County-Enforcement Heterogeneity DiD - Restricted-Use CHIS Adults

Outcome	Coefficient	Beta	SE	p
Medi-Cal	Naturalized x Post (low-enf baseline)	+0.006	0.012	0.629
Medi-Cal	Green-card x Post (low-enf baseline)	+0.039	0.025	0.124
Medi-Cal	No-green-card x Post (low-enf baseline)	+0.125	0.022	<0.001
Medi-Cal	High-enf x Naturalized x Post	-0.002	0.008	0.785
Medi-Cal	High-enf x Green-card x Post	-0.032	0.018	0.074
Medi-Cal	High-enf x No-green-card x Post	-0.045	0.019	0.021
Usual source	Naturalized x Post (low-enf baseline)	+0.001	0.011	0.930
Usual source	Green-card x Post (low-enf baseline)	+0.076	0.024	0.002
Usual source	No-green-card x Post (low-enf baseline)	+0.130	0.024	<0.001
Usual source	High-enf x Naturalized x Post	+0.009	0.009	0.331
Usual source	High-enf x Green-card x Post	-0.060	0.017	<0.001
Usual source	High-enf x No-green-card x Post	-0.084	0.019	<0.001
ED visit	Naturalized x Post (low-enf baseline)	+0.052	0.011	<0.001
ED visit	Green-card x Post (low-enf baseline)	+0.066	0.022	0.004
ED visit	No-green-	+0.077	0.020	<0.001

Outcome	Coefficient	Beta	SE	p
ED visit	card x Post (low-enf baseline)			
ED visit	High-enf x Naturalized x Post	-0.004	0.008	0.601
ED visit	High-enf x Green-card x Post	-0.021	0.014	0.135
ED visit	High-enf x No-green- card x Post	-0.032	0.016	0.049
Forgone care	Naturalized x Post (low-enf baseline)	-0.012	0.008	0.138
Forgone care	Green-card x Post (low-enf baseline)	-0.007	0.011	0.533
Forgone care	No-green- card x Post (low-enf baseline)	-0.001	0.017	0.933
Forgone care	High-enf x Naturalized x Post	+0.001	0.006	0.927
Forgone care	High-enf x Green-card x Post	-0.016	0.008	0.058
Forgone care	High-enf x No-green- card x Post	+0.019	0.013	0.162

Notes: Restricted-use CHIS adult sample, 2015-2024 (N = 220,925; sample matches all four outcomes). Specification: four-level CITIZEN1 multi-group DiD with high-enforcement-county interactions, controlling for age, sex, race, below-200% FPL, survey-year fixed effects, and CITIZEN1 fixed effects. U.S.-born citizens omitted. The low-enforcement baseline corresponds to the San Francisco AOR (48 California counties); high-enforcement extra coefficients identify the differential post-period response in the Los Angeles + San Diego AORs (10 counties; 100 county-years, ~17% of the merged county-year set; 480 SF-AOR county-years, ~83%). High-enforcement is an AOR-level above-median per-capita arrest indicator constructed from the Deportation Data Project (CC-0; data accessed 2026-04-24). Because public AOR-level data begin in fiscal year 2022, 2015-2021 county-years use the AOR's 2022 value as the earliest

publicly observable enforcement-intensity rank, so the indicator is essentially time-invariant within the AOR-pair partition during the 2018-2024 paper window. Substantive interpretation is a Southern-versus-Northern California enforcement-intensity contrast on the level of the post-2018 response, not a within-AOR county-level gradient and not a time-varying enforcement-shock design. The within-noncitizen-group post coefficients (low-enforcement baseline) are dominated by California's full-scope Medi-Cal expansions for undocumented adults during the paper window (January 2020 ages 19-25, May 2022 ages 50+, January 2024 ages 26-49); the high-enforcement-extra interactions identify a consistent differential (adults without green cards in higher-enforcement AORs gained less coverage and access during the expansion calendar than adults without green cards in the lower-enforcement AOR), most clearly on the access margin (usual source of care, -8.4 pp, $p < 10^{-4}$). Variance estimated via 80 CHIS replicate jackknife weights using the design-supplied multiplier. Author's calculations from restricted-use CHIS adult output returned by the UCLA DAC.

Appendix Exhibit A19: CHIS Two-Window Decomposition - Medi-Cal, Four-Level CITIZEN1

Sample	Coefficient	Beta	SE	p
Full sample (N = 220,925)	Naturalized x Chilling window (2019-2021)	-0.001	0.010	0.954
Full sample	Green-card x Chilling window	+0.017	0.024	0.491
Full sample	No-green- card x Chilling window	+0.054	0.020	0.008
Full sample	Naturalized x Expansion window (2022-2024)	+0.009	0.012	0.425
Full sample	Green-card x Expansion window	+0.022	0.023	0.346
Full sample	No-green- card x Expansion	+0.134	0.019	<0.001

Sample	Coefficient	Beta	SE	p
Full sample	window No-green-card: chilling minus expansion	-0.080	0.018	<0.001
Age 26-49 (N = 64,867)	No-green-card x Chilling window	+0.071	0.024	0.004
Age 26-49	No-green-card x Expansion window	+0.108	0.022	<0.001
Age 26-49	No-green-card: chilling minus expansion	-0.037	0.021	0.082

Notes: Restricted-use CHIS adult sample, 2015-2024, four-level CITIZEN1 decomposition with U.S.-born citizens as the reference group. The “chilling window” interacts group indicators with survey years 2019-2021 (the federal public charge period before California’s major adult full-scope expansions for ages 50+ and 26-49); the “expansion window” covers 2022-2024. The positive no-green-card Medi-Cal coefficients in both windows, and the significantly larger expansion-window coefficient, show that the pooled post-2018 gain among adults without green cards is dominated by California’s full-scope Medi-Cal expansion calendar rather than reflecting an absence of federal chilling pressure (Section 5.7). Author’s calculations from restricted-use CHIS adult output returned by the UCLA DAC.

Appendix Exhibit A20: CHIS Age-Stratified Four-Level DiD - Medi-Cal

Age band	No-green-card x Post	SE	p	N	CA full- scope expansion for this band
19-25	+0.102	0.062	0.106	12,831	January 2020
26-49	+0.090	0.020	<0.001	64,867	January 2024
50-64	+0.172	0.044	<0.001	65,102	May 2022
65+	-0.040	0.126	0.749	76,014	Medicare- eligible (no CA expansion)

Age band	No-green-card x Post	SE	p	N	CA full- scope expansion for this band mechanis m)
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Notes: Restricted-use CHIS adult sample, 2015-2024, four-level CITIZEN1 multi-group DiD estimated separately by age band; U.S.-born citizens are the reference group and the table reports the no-green-card x Post coefficient for Medi-Cal enrollment. The age-band pattern lines up with California's full-scope Medi-Cal expansion calendar for undocumented adults, confirming that the positive pooled coefficient is expansion-driven (Section 5.7). Corresponding figures appear as Appendix Figures A17-A18. Author's calculations from restricted-use CHIS adult output returned by the UCLA DAC.

Appendix Exhibit A21: Administrative Vulnerability - CHIS Documentation Demands

(Immigrant-Only Universe)

Specificatio n	Estimate	SE	p	N
(1) Noncitizen vs. naturaliz ed, year FE only	+0.092	0.008	<0.001	33,464
(2) + age, sex, below-200 % FPL	+0.055	0.008	<10⁻¹⁰	33,464
(3) + race/ethnicit y dummies	+0.057	0.009	<10⁻¹⁰	33,427
Reference- group mean (naturalized)	0.105	-	-	24,085

Notes: CHIS public-use adult sample, 2019-2024, restricted to respondents not born in the United States - the survey universe for the AL19V2 documentation-demand item. Each row is a weighted linear-probability regression of being asked for a Social Security number or citizenship documentation when seeking coverage on a noncitizen indicator (naturalized = reference), with

year fixed effects and the listed controls. Heteroskedasticity-robust standard errors. A specification that compared both groups against a U.S.-born baseline of structural zeros (the question is filtered to immigrants) would mechanically inflate the gradient and is not reported. The corrected within-immigrant-universe gradient is 5-9 percentage points depending on covariate set: noncitizens face that much more documentation scrutiny than naturalized citizens at the same enrollment-system entry points (Section 5.9).

Appendix Exhibit A22: CHIS California Children in Mixed-Status Families (Below 200% FPL)

Outcome	Beta	SE	p	N
Medi-Cal	+0.00	0.031	0.806	9,607
	8			
Any public coverage	+0.00	0.031	0.896	9,607
	4			
Any coverage	+0.02	0.013	0.060	9,607
	4			
Uninsured	-0.024	0.013	0.060	9,607
Private coverage	+0.02	0.026	0.439	9,607
	0			

Notes: CHIS child sample, California children below 200% FPL in mixed-status versus citizen-only families, pooled DiD around the September 2018 announcement. All coverage outcomes are null. California expanded full-scope Medi-Cal to undocumented children in May 2016, before the treatment window, and the CHIS California child sample (N = 9,607) is small relative to the national ACS child sample. The null is consistent with California having neutralized the federal chilling effect on children's coverage through a more inclusive state-funded eligibility framework, not with absence of national chilling (Section 6.5). Author's calculations from CHIS child microdata.

Appendix Figures

Figure A1. Uninsured event study.

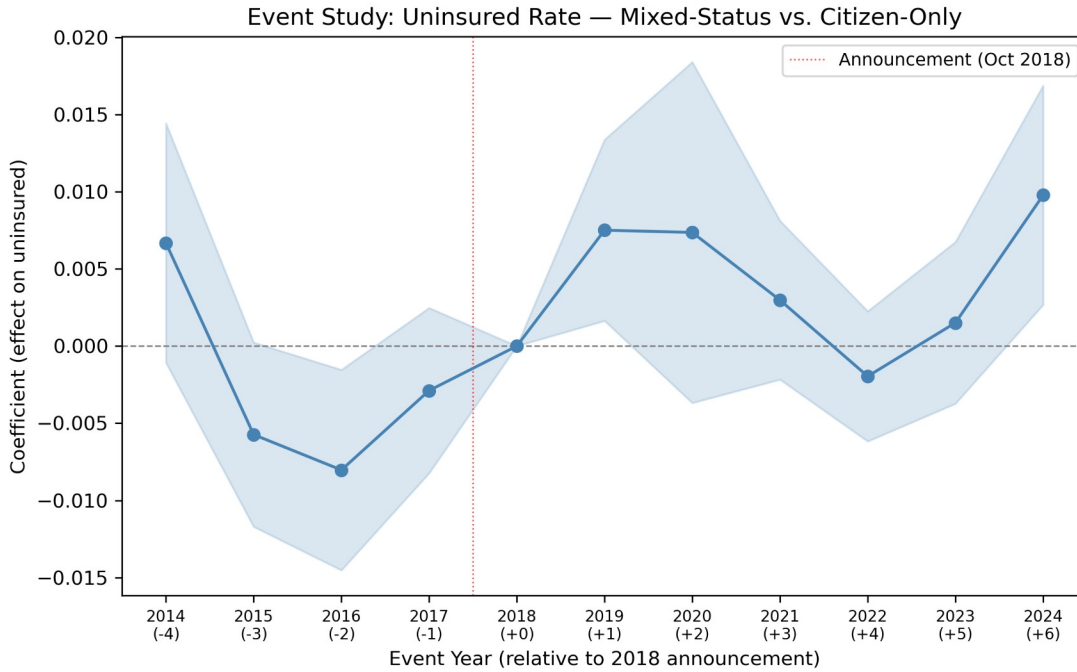


Figure A1

Figure A2. Any-coverage event study.

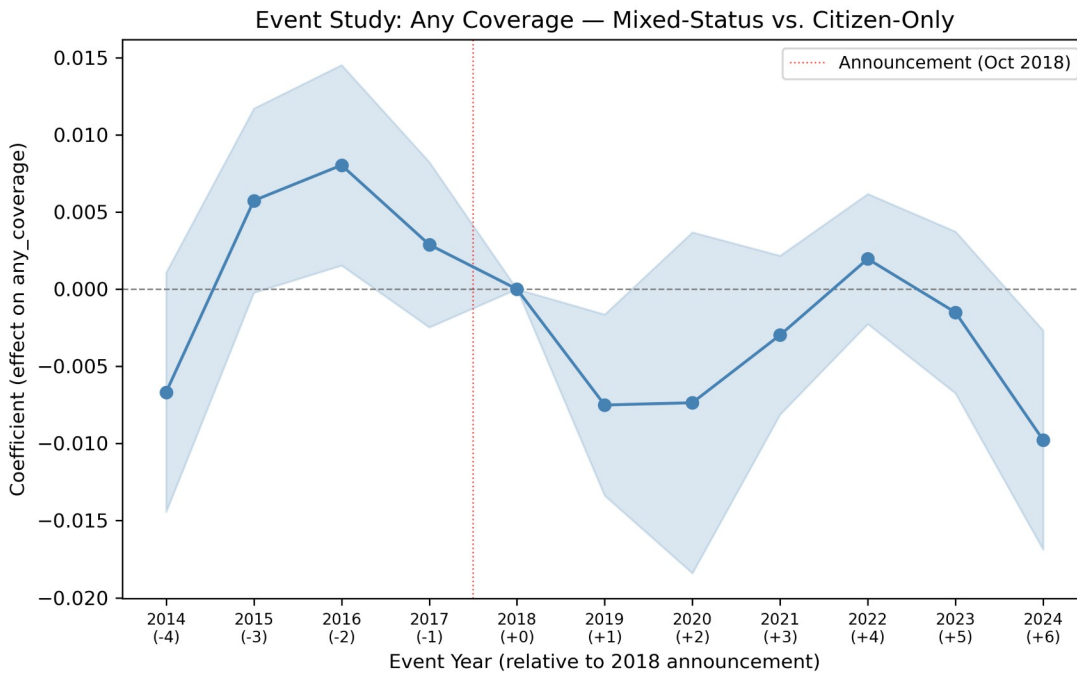


Figure A2

Figure A3. Medicaid event study, full sample.

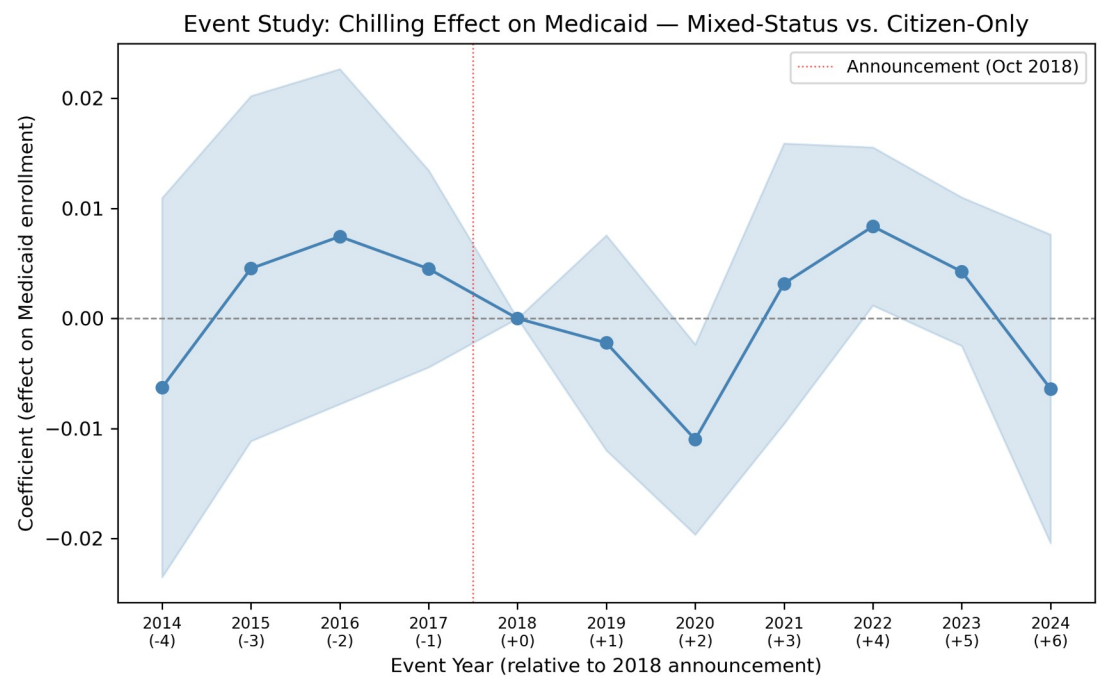


Figure A3

Figure A4. Placebo event study, citizen-only families.

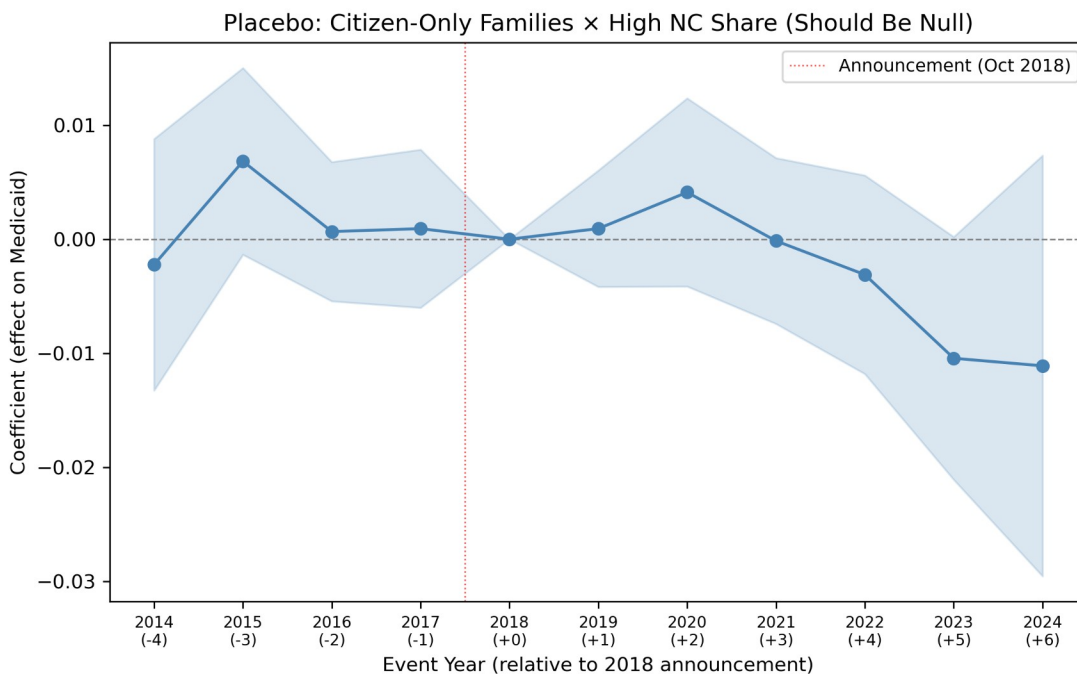


Figure A4

Figure A5. ACS covariate balance.

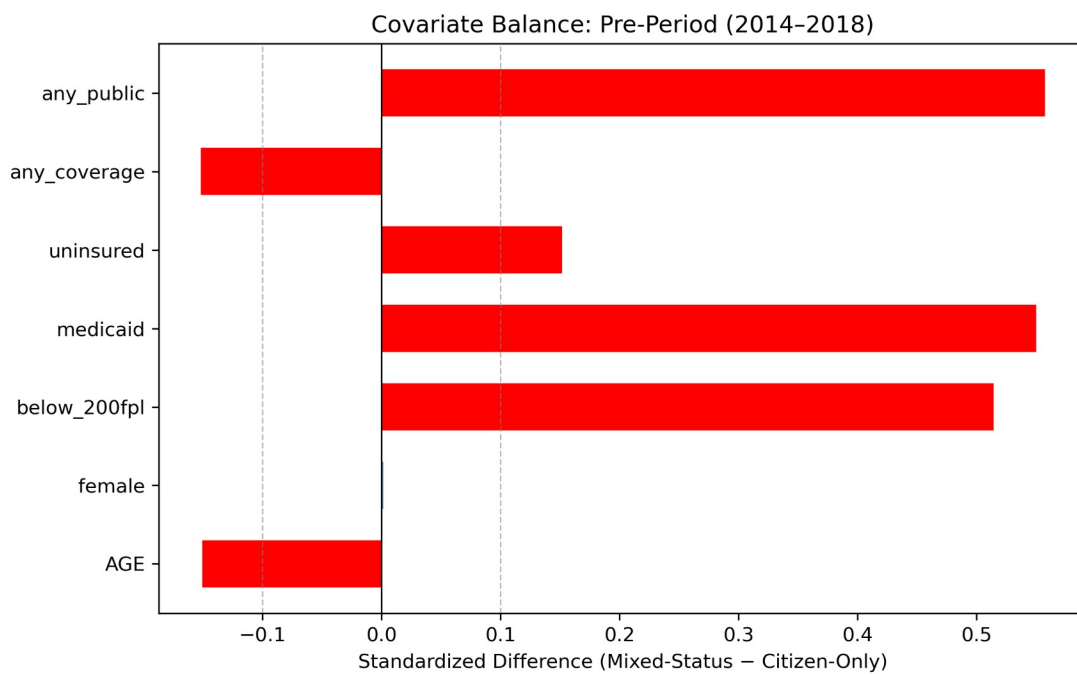


Figure A5

Figure A6. CHIS covariate balance.

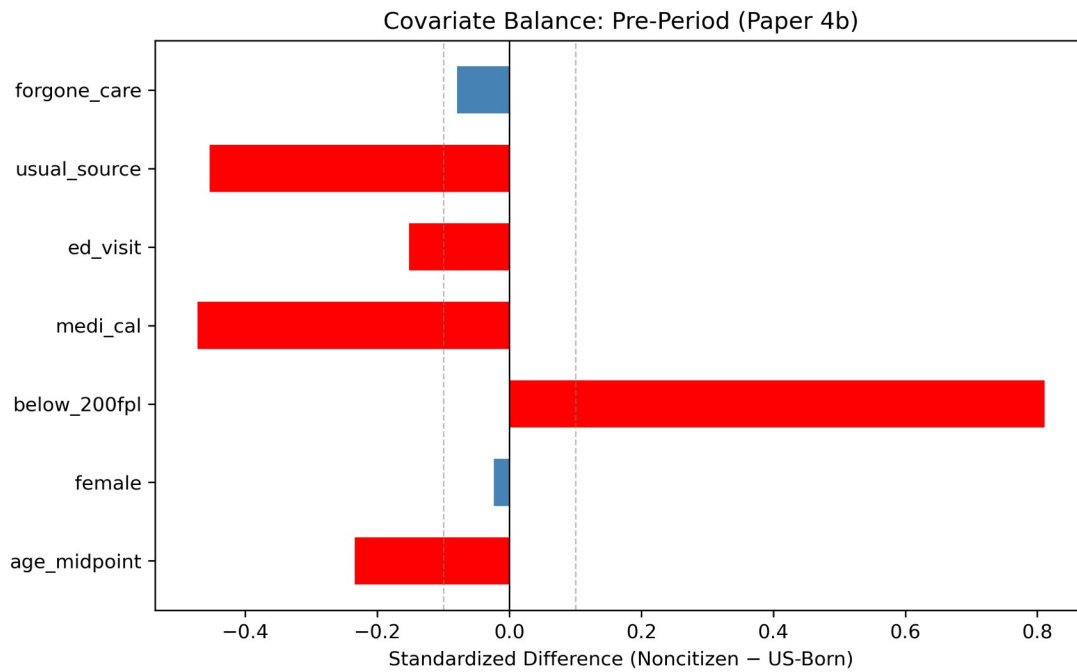


Figure A6

Figure A7. Medicaid race/ethnicity heterogeneity.

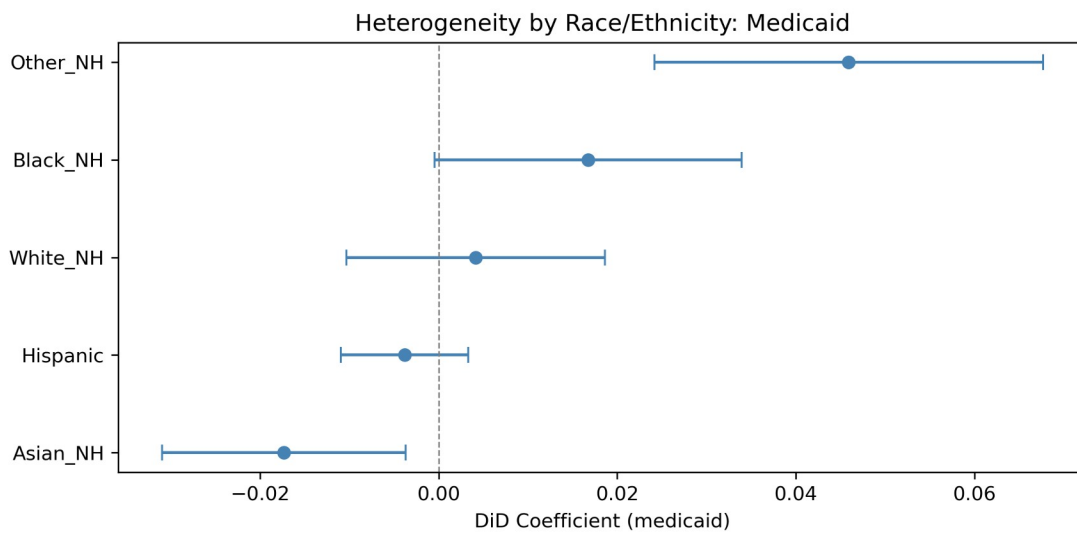


Figure A7

Figure A8. Uninsured race/ethnicity heterogeneity.

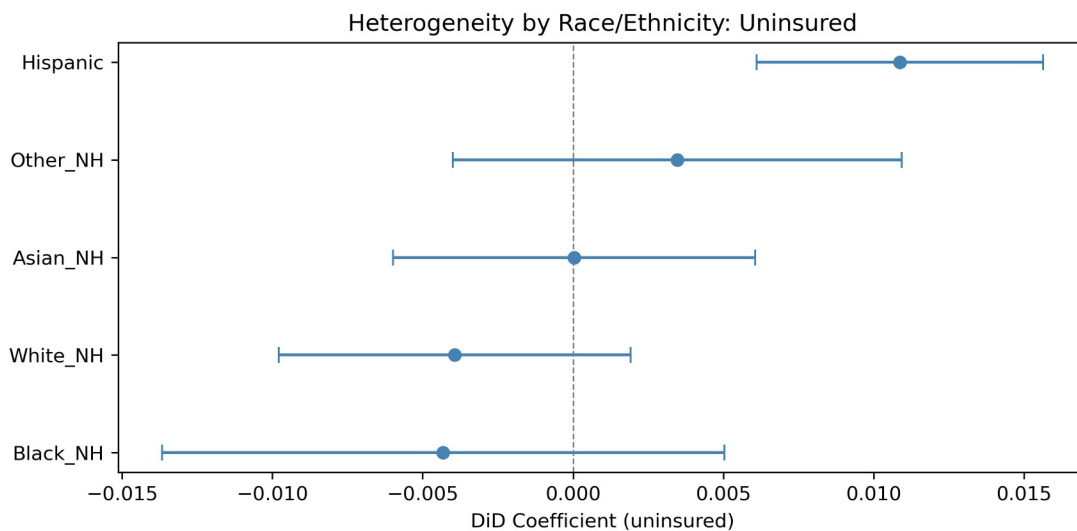


Figure A8

Figure A9. CHIS event-study panels.

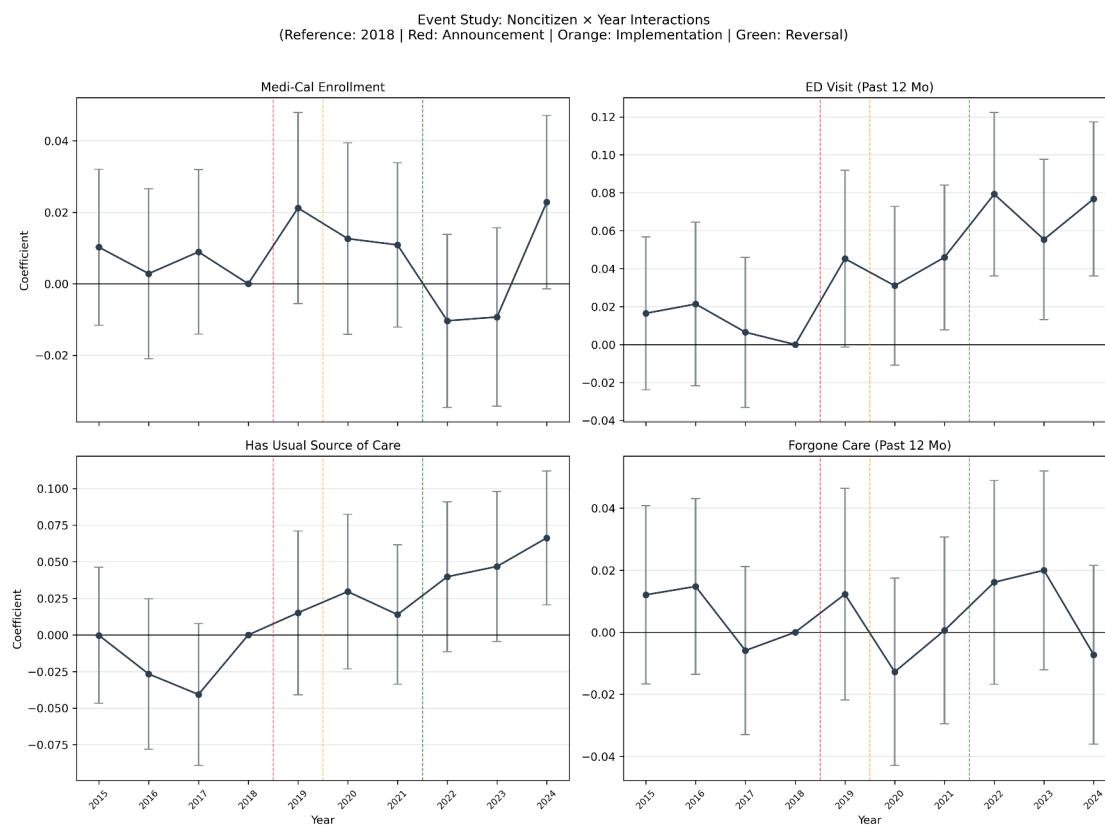


Figure A9

Figure A10. Naturalized-citizen placebo event study.

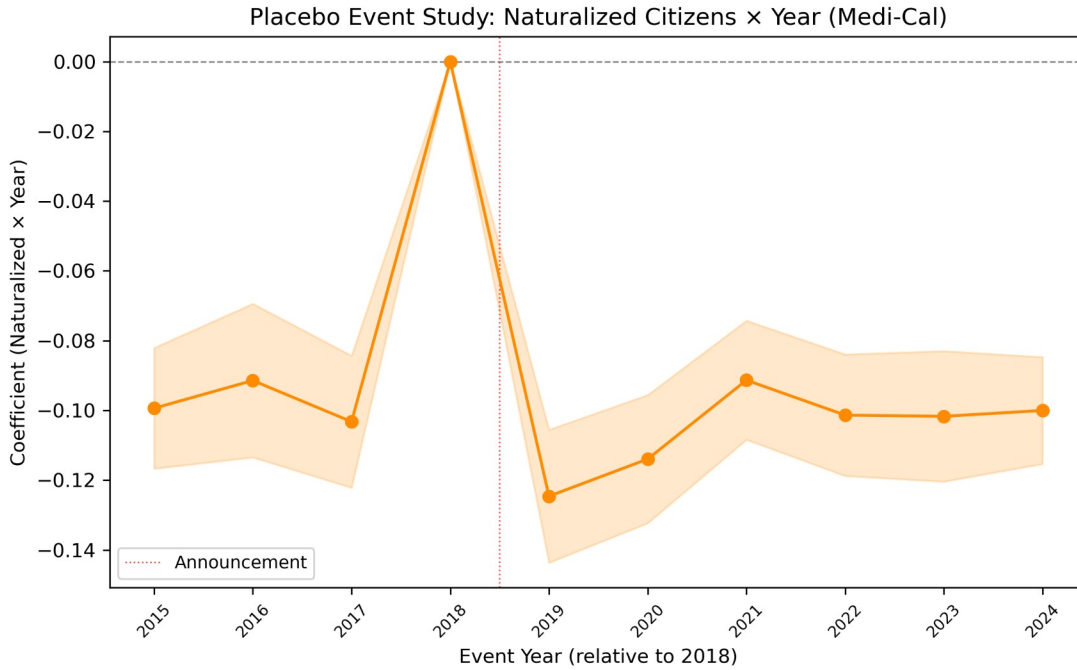


Figure A10

Figure A11. Early-timing Medicaid event study (reference 2016).

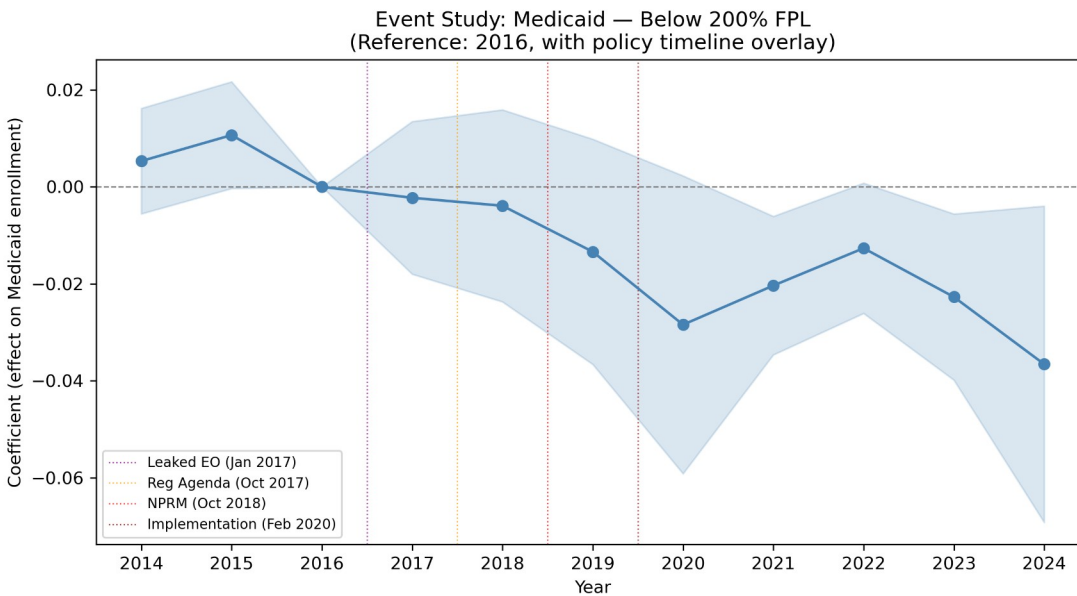


Figure A11

Figure A12. Raw outcome trends, both samples.

Raw Outcome Trends: Mixed-Status vs. Citizen-Only Families

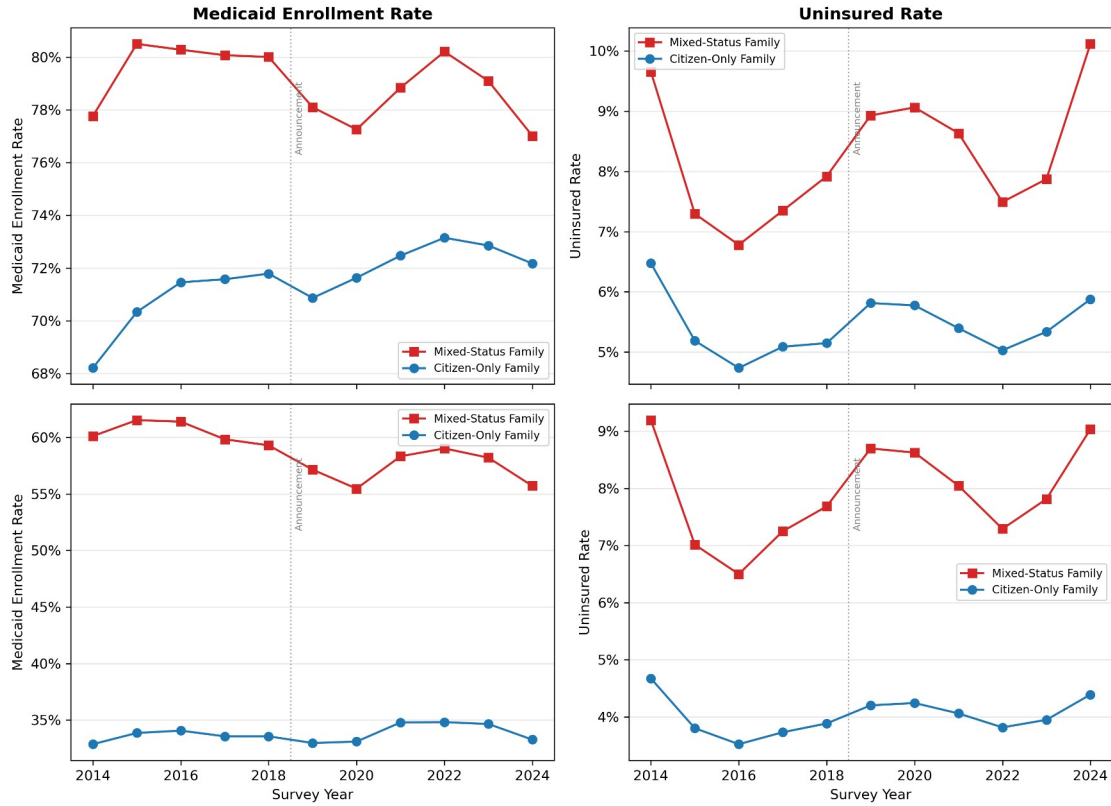


Figure A12

Figure A13. Multi-outcome event study, levels.

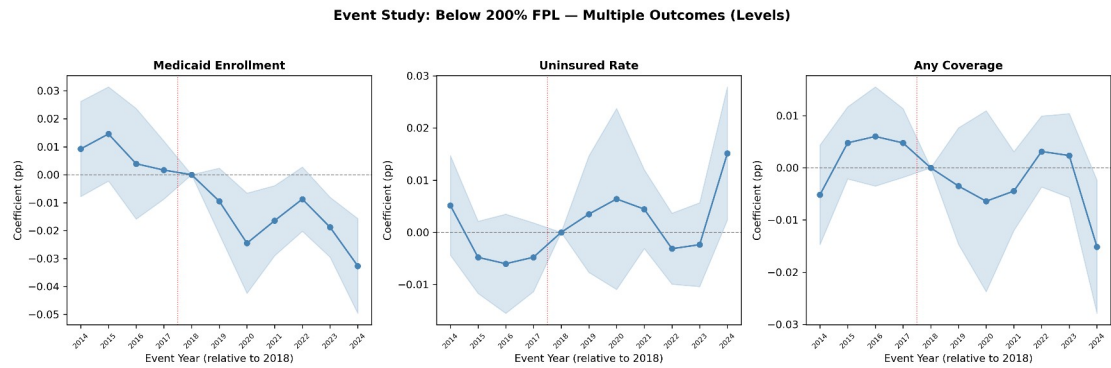


Figure A13

Figure A14. Multi-outcome event study, percent change.

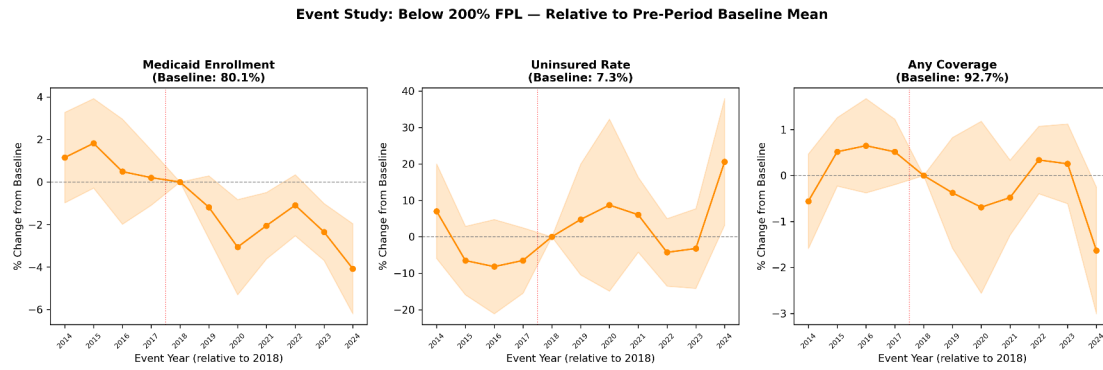


Figure A14

Figure A15. Pairwise CHS Medi-Cal event studies.

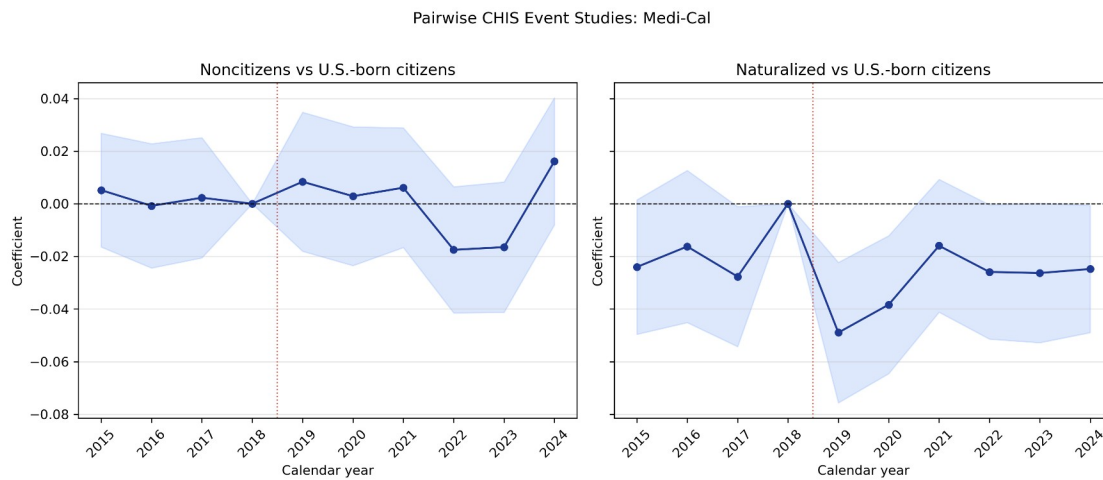


Figure A15

Figure A16. State-by-year FE Medicaid event study.

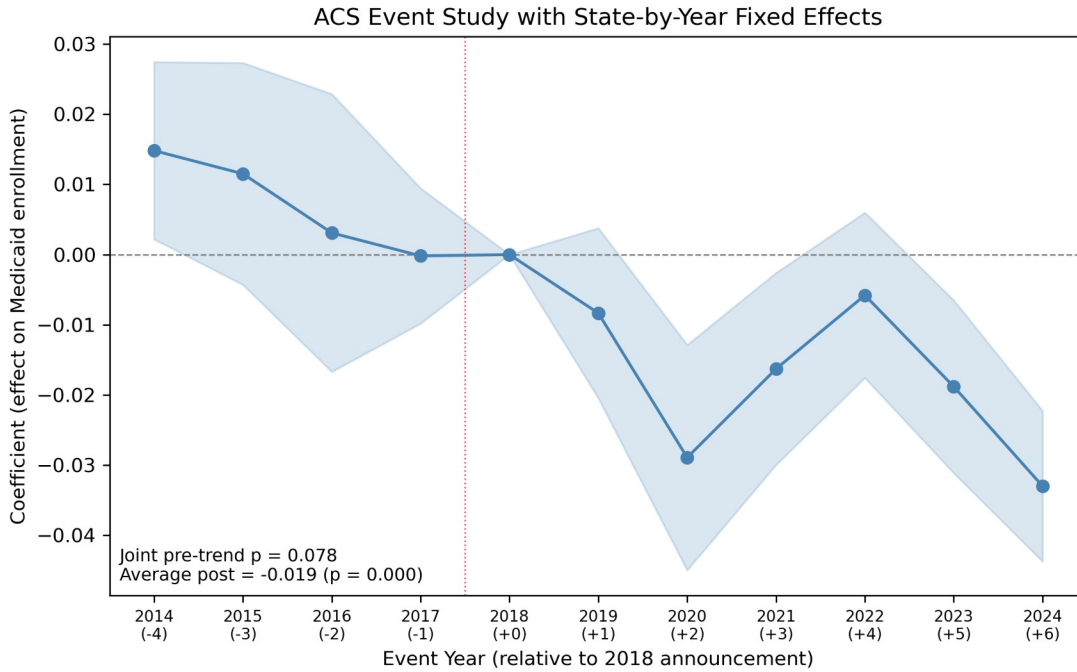


Figure A16

Figure A17. CHIS age-stratified Medi-Cal effects (four-level CITIZEN1).

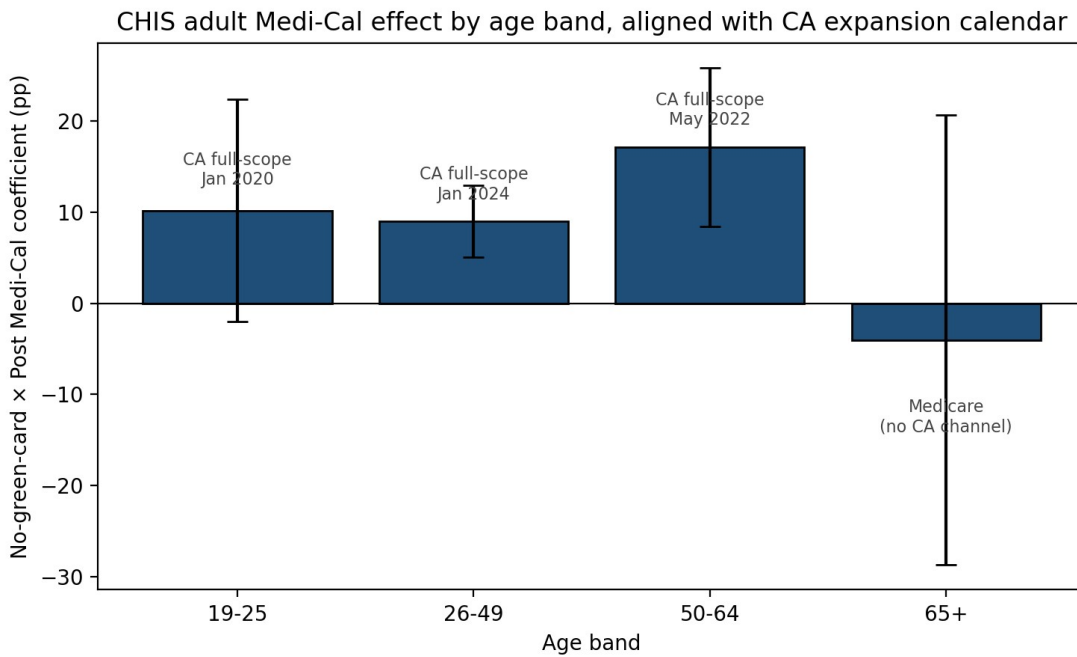


Figure A17

Figure A18. CHIS age 26-49 Medi-Cal event study.

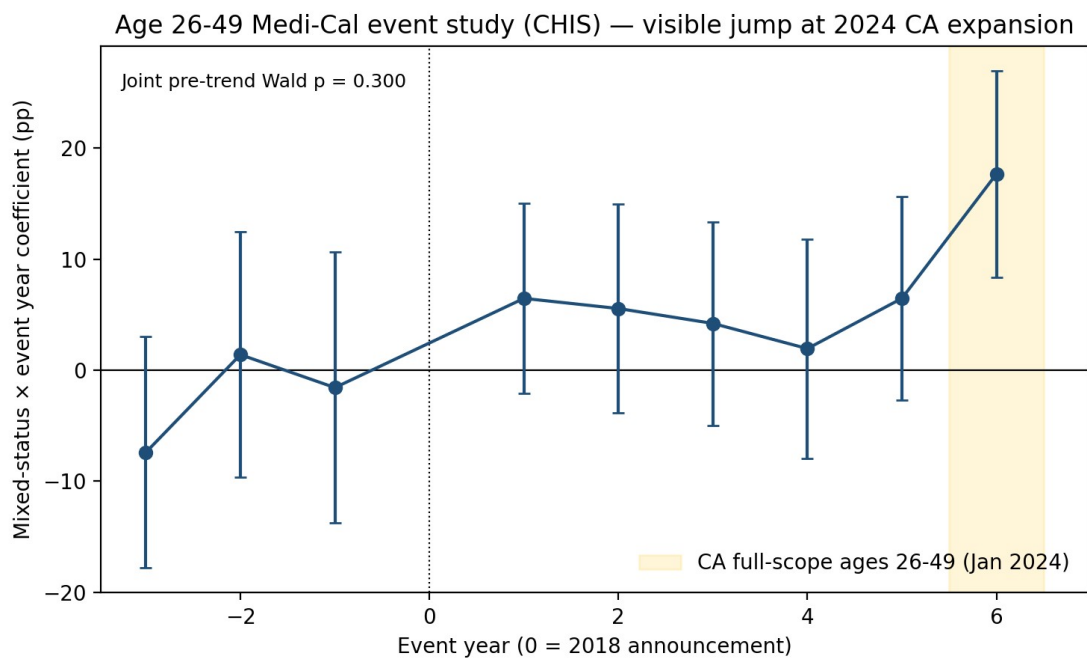


Figure A18